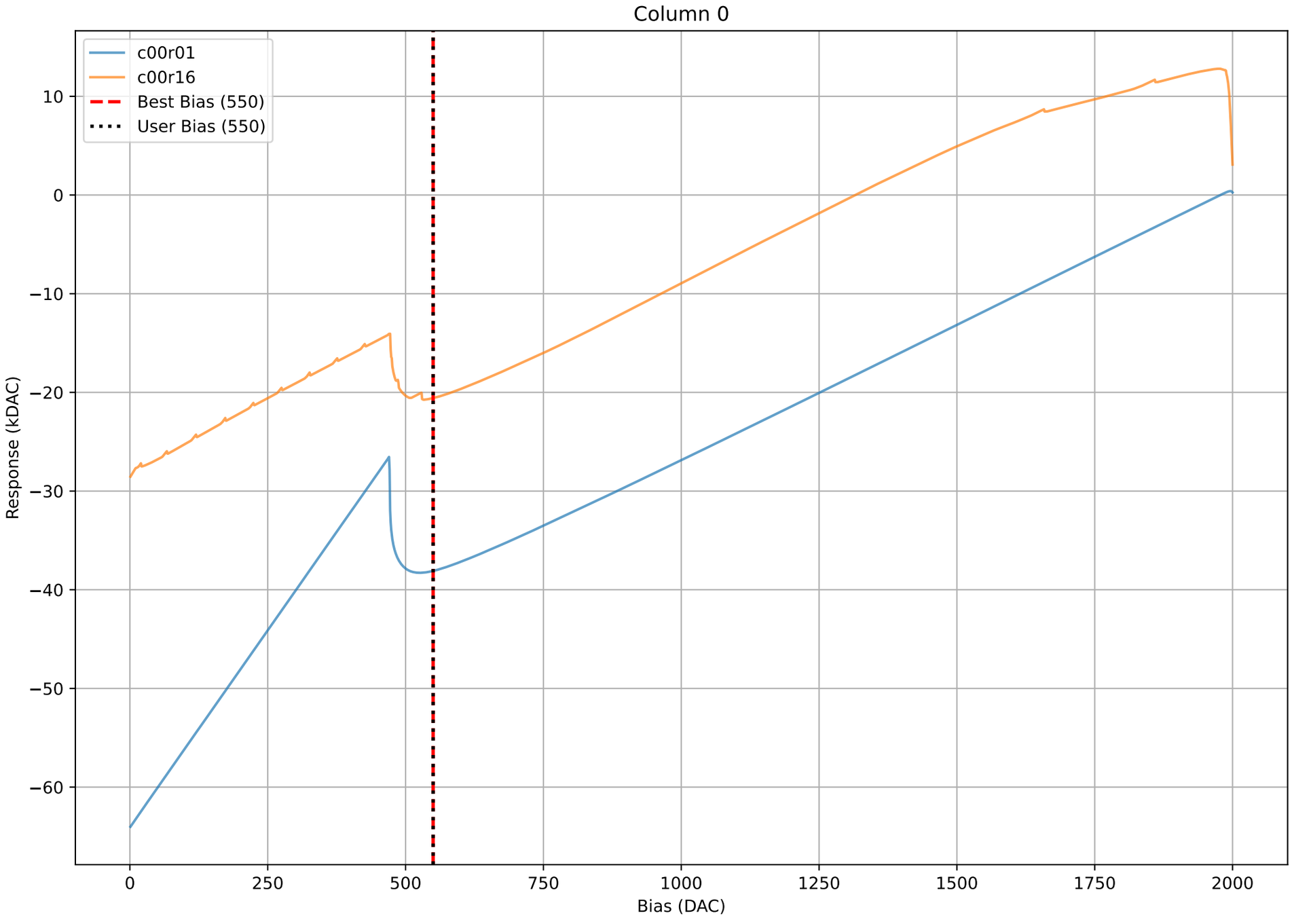
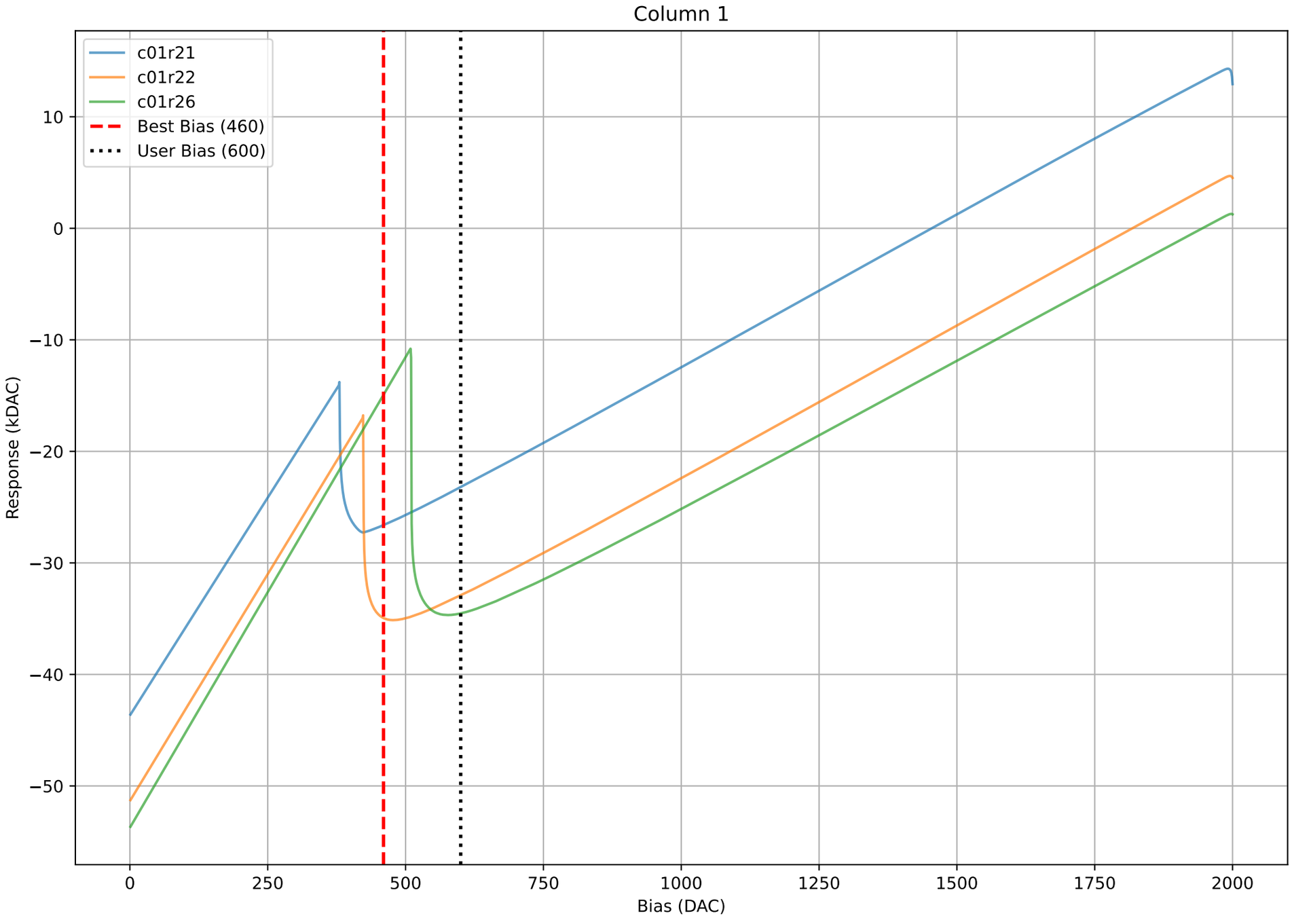
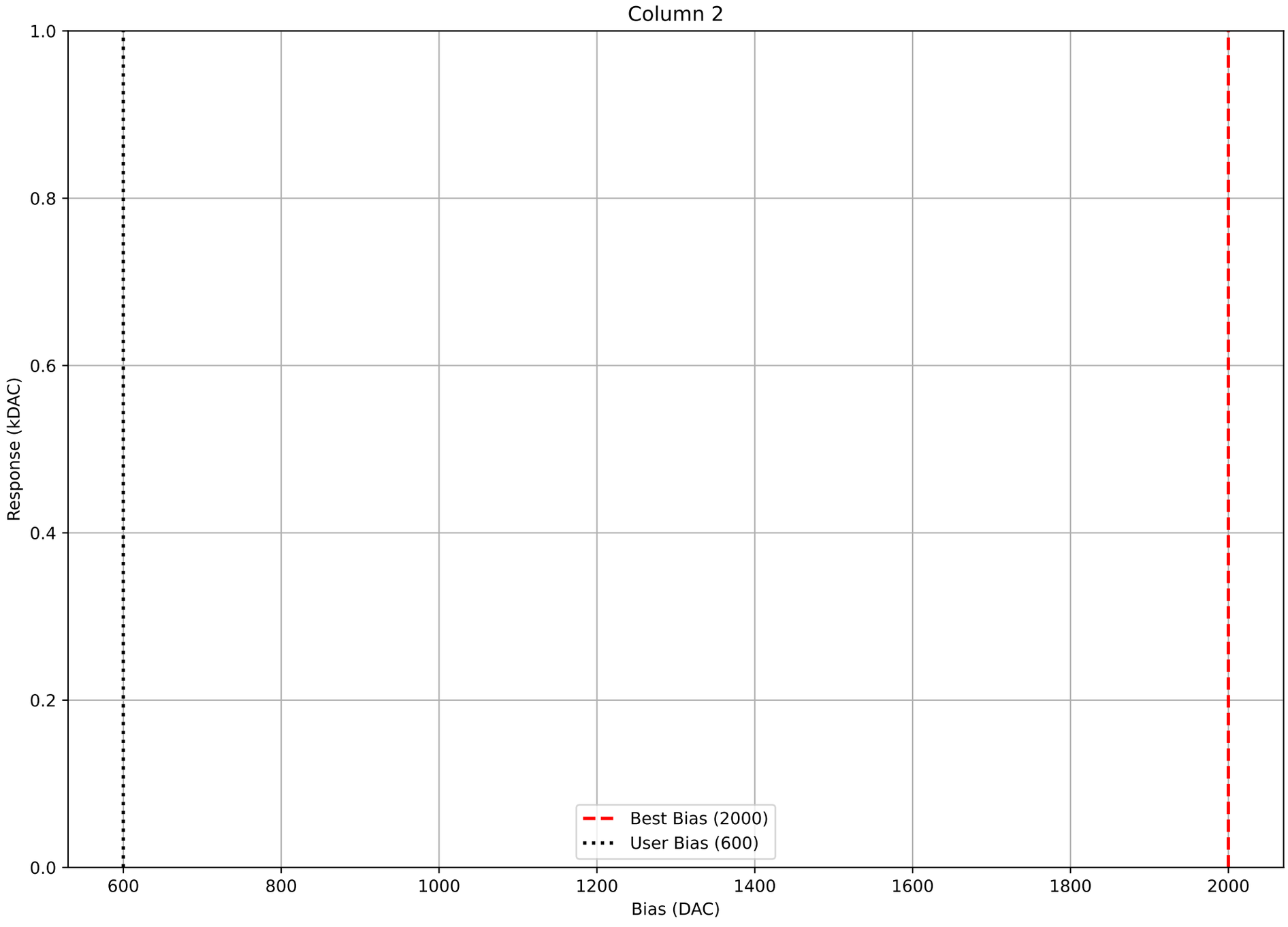
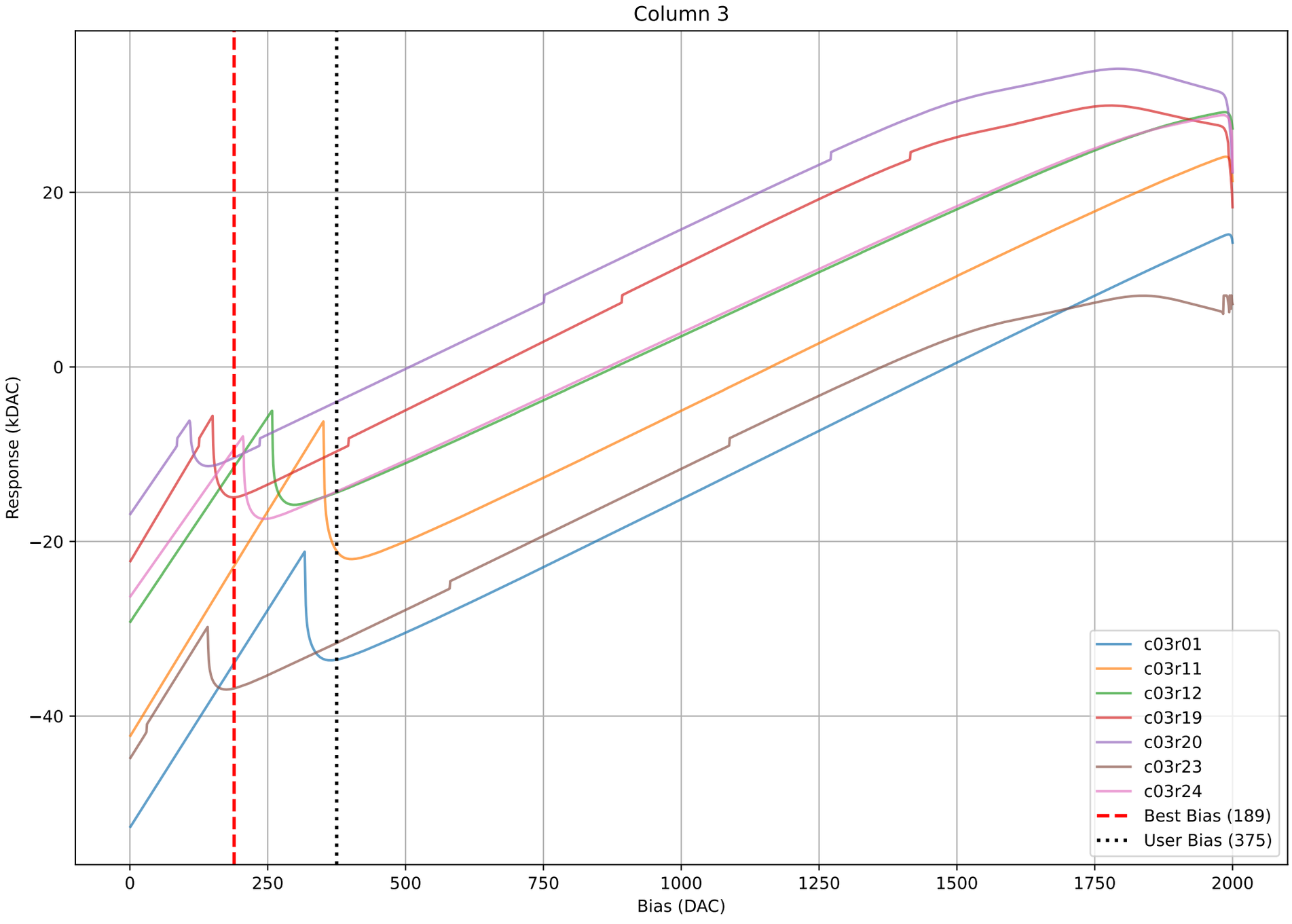
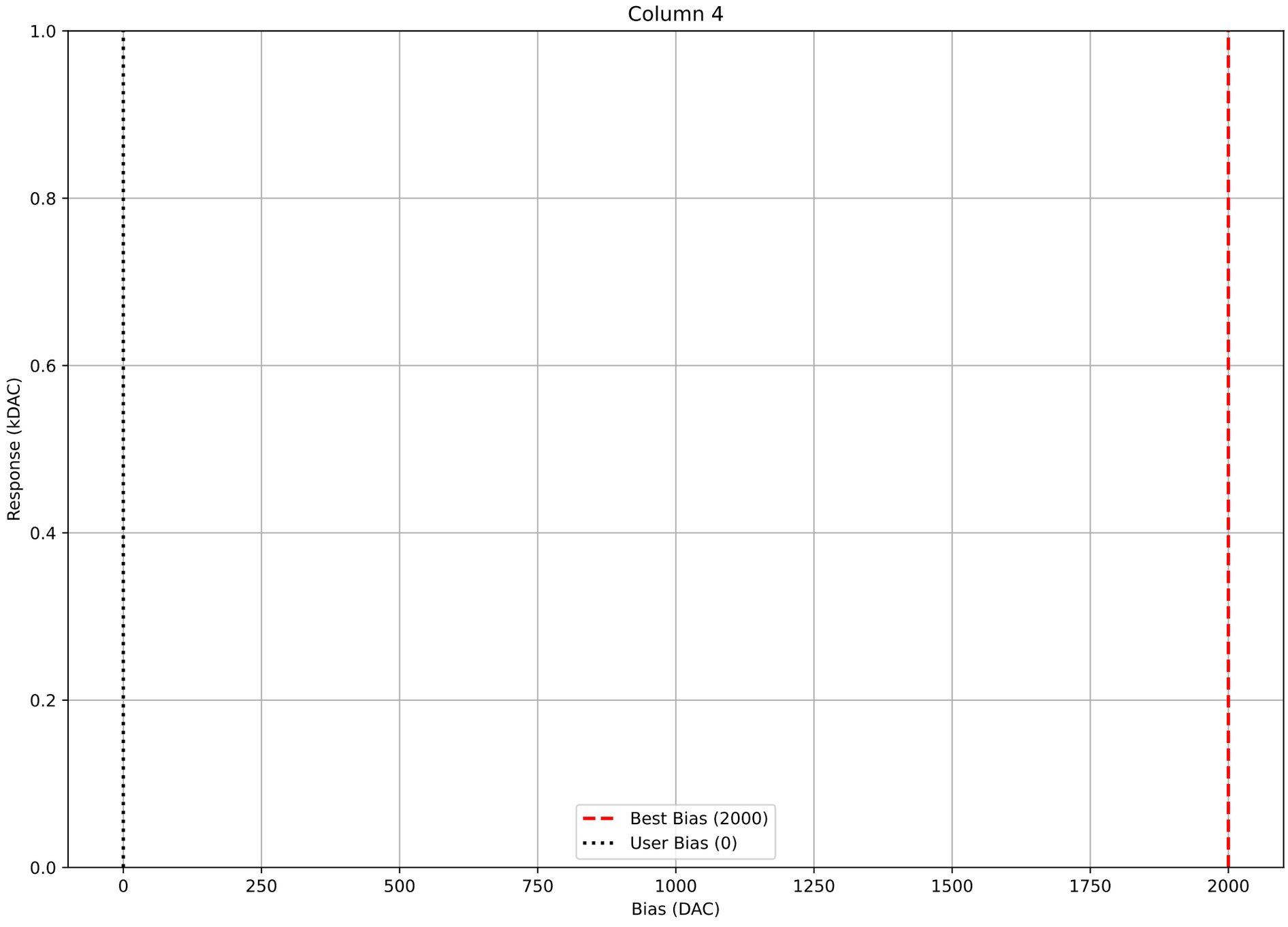
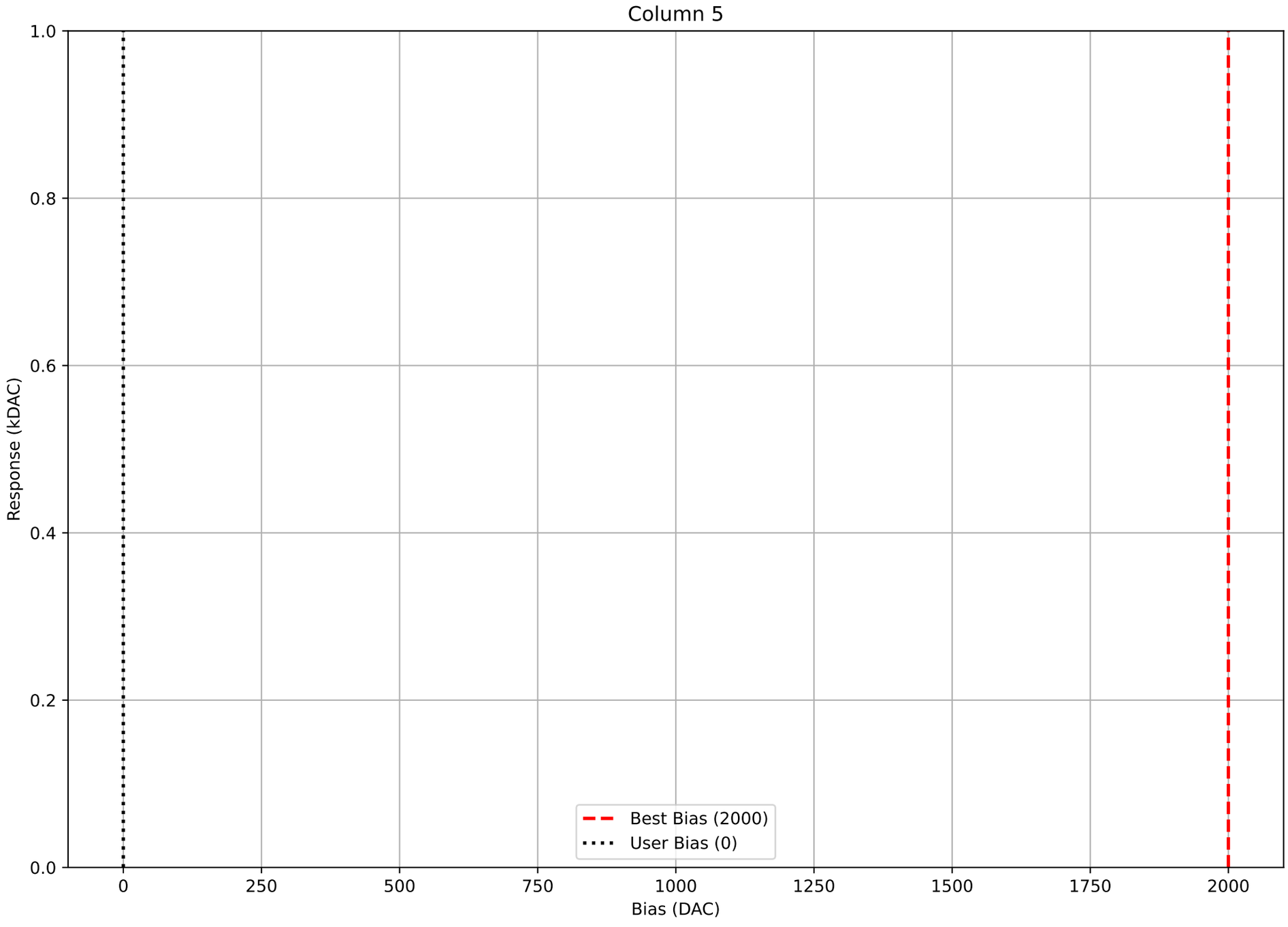

Column 0
c00r01
c00r16
Best Bias (550)
User Bias (550)
Response (kDAC)
10
0
−10
−20
−30
−40
−50
−60
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

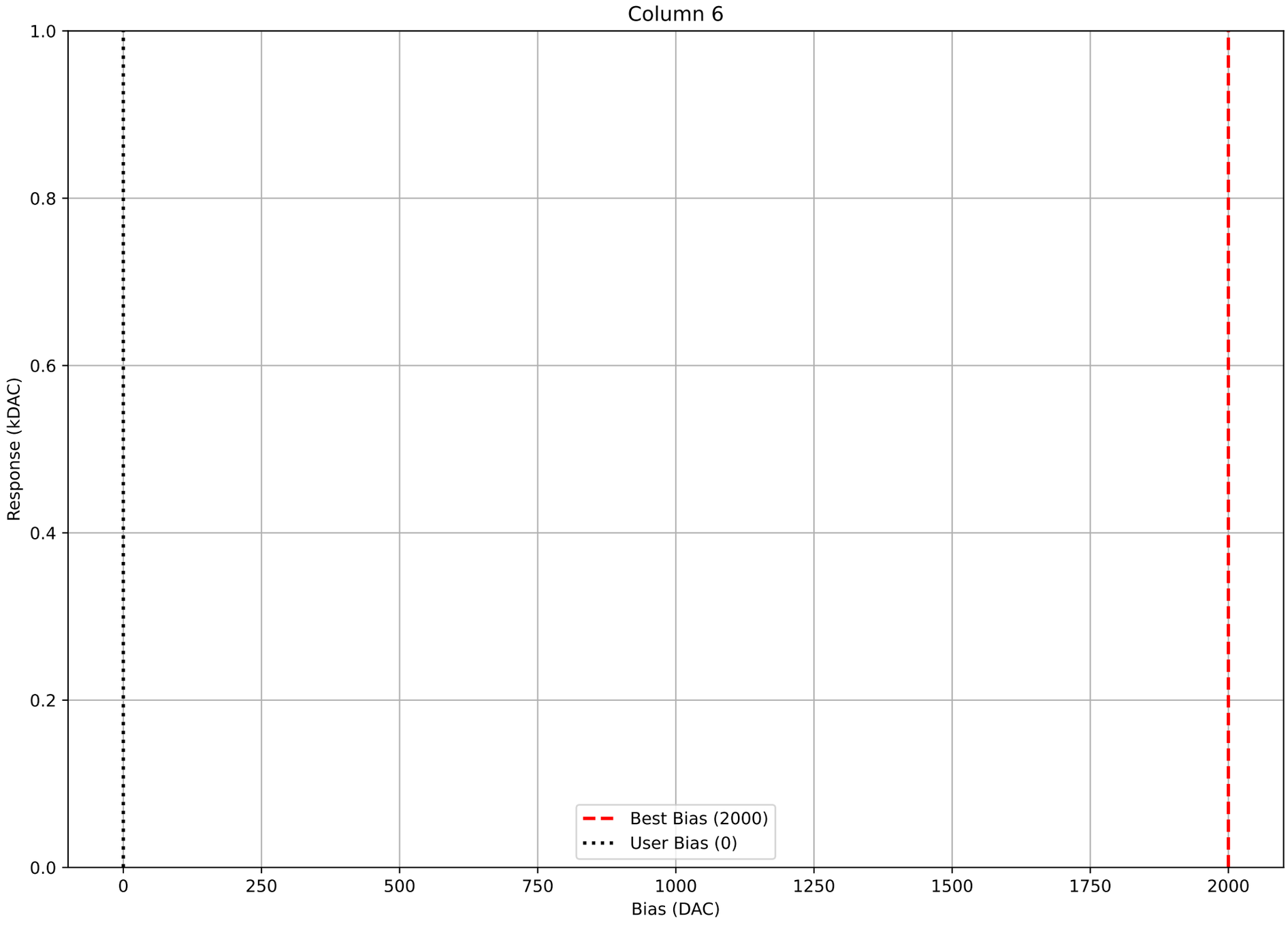

Column 1
c01r21
c01r22
c01r26
Best Bias (460)
User Bias (600)
Response (kDAC)
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

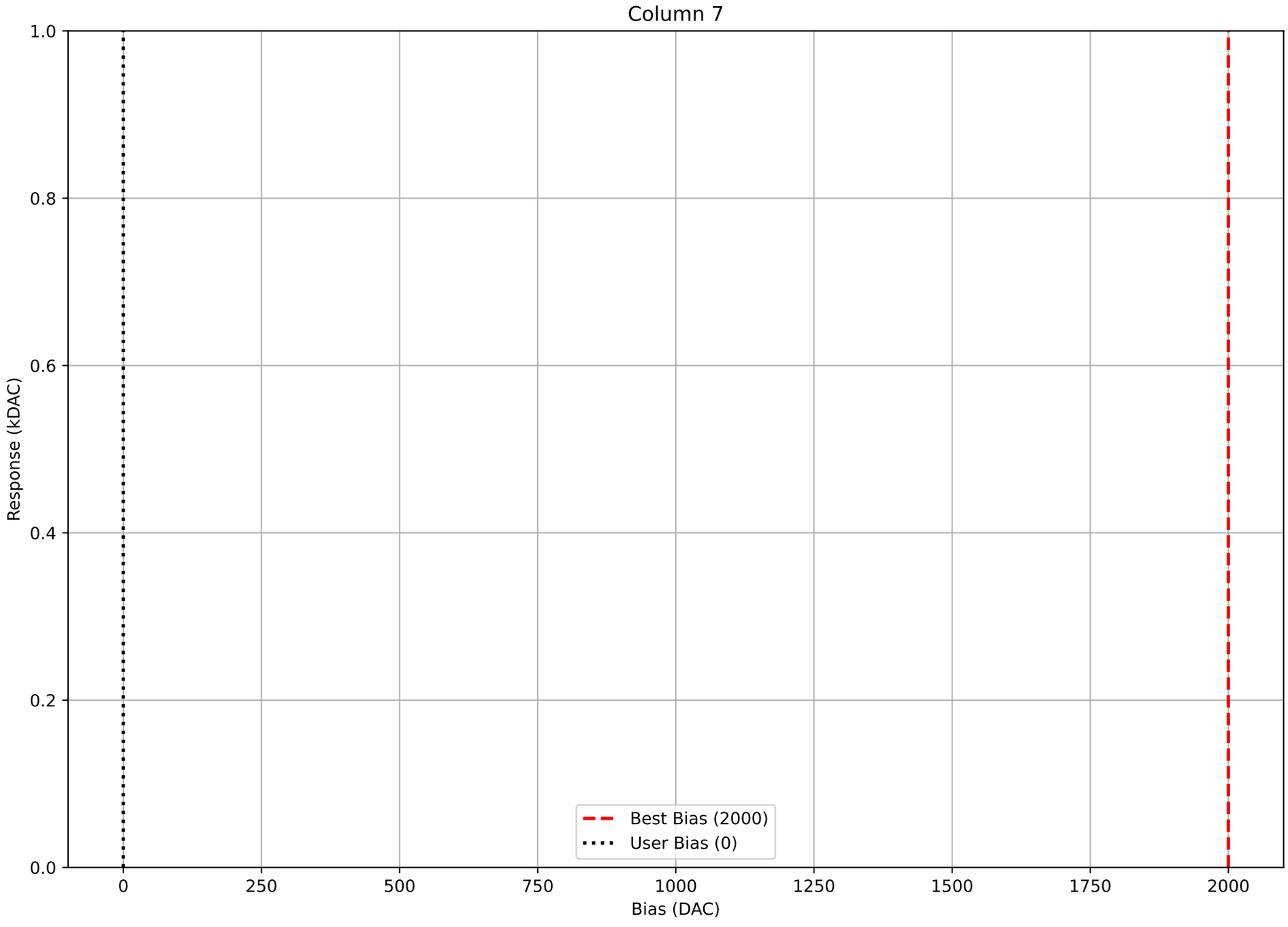

Column 2
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
600
800
1000
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (600)

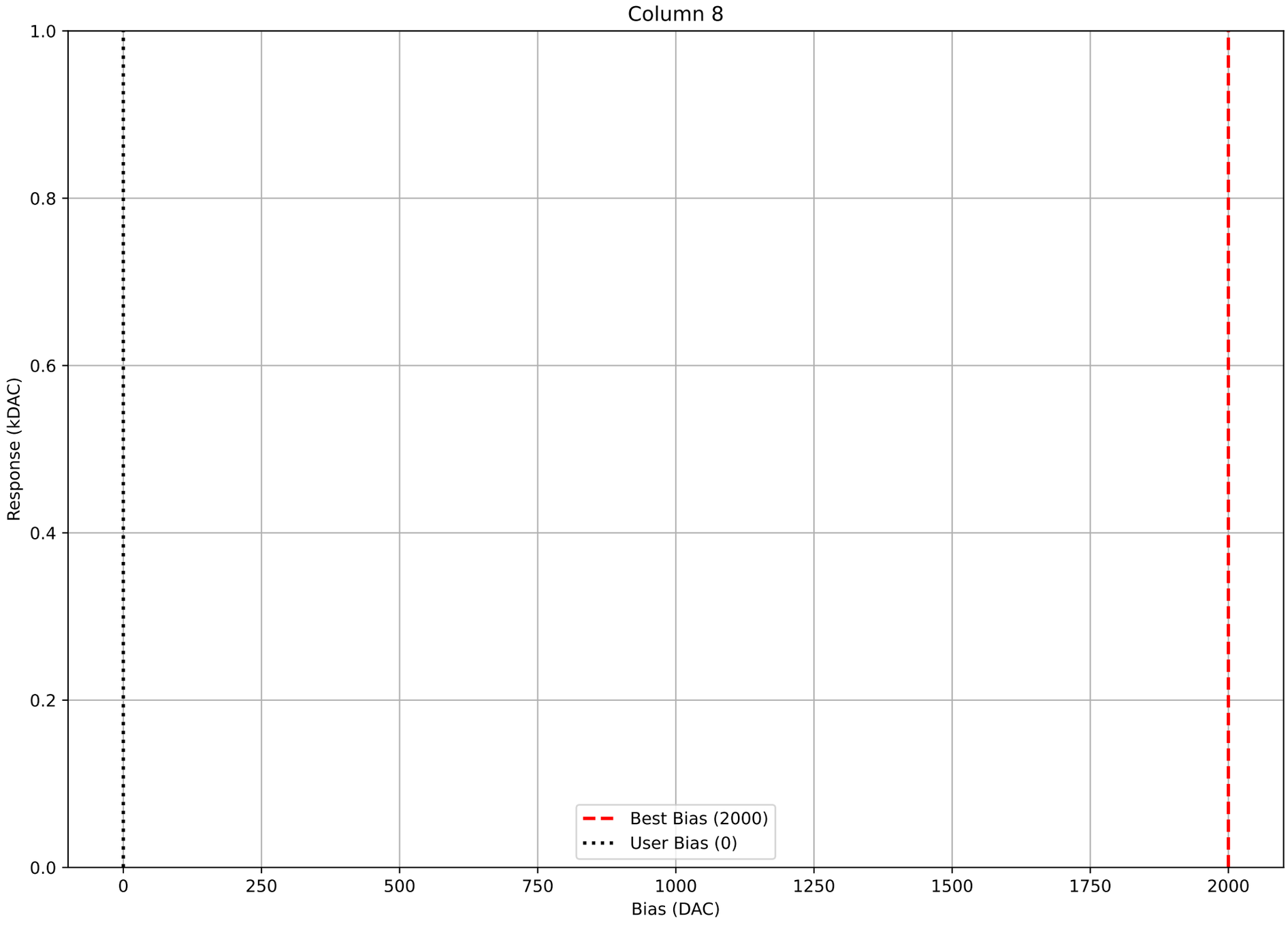

Column 3
Response (kDAC)
20
0
−20
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c03r01
c03r11
c03r12
c03r19
c03r20
c03r23
c03r24
Best Bias (189)
User Bias (375)

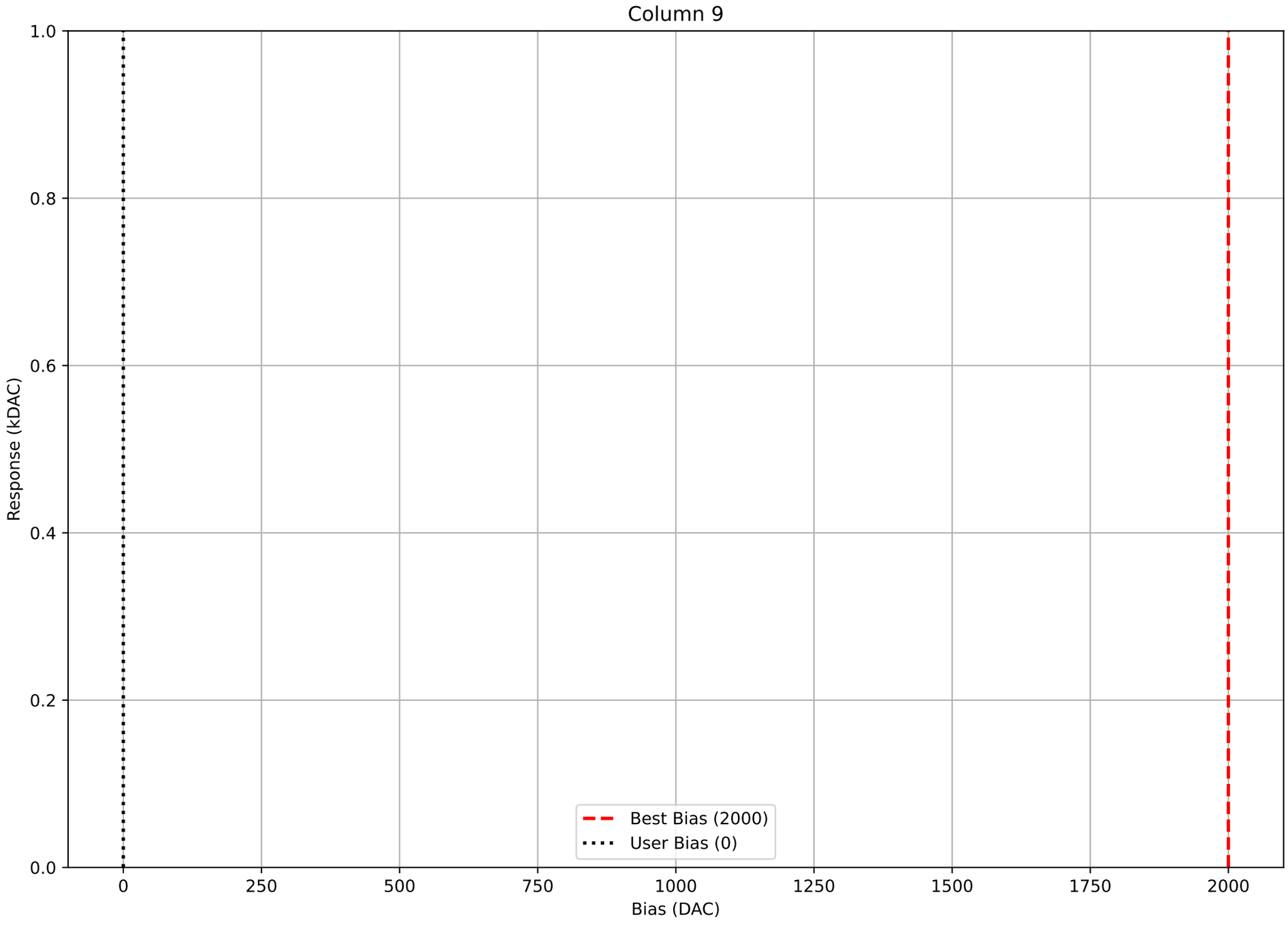

Column 4
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

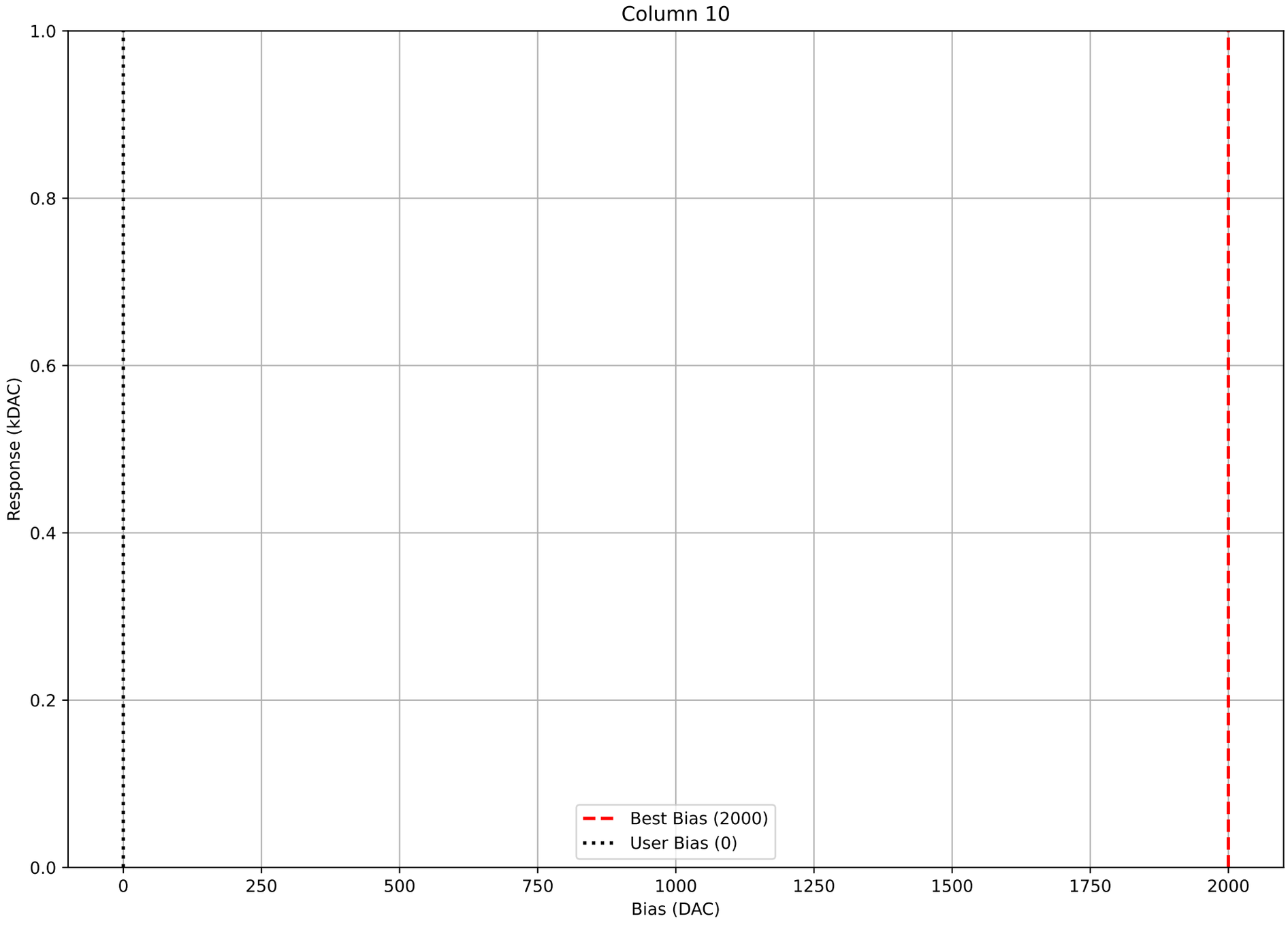

Column 5
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

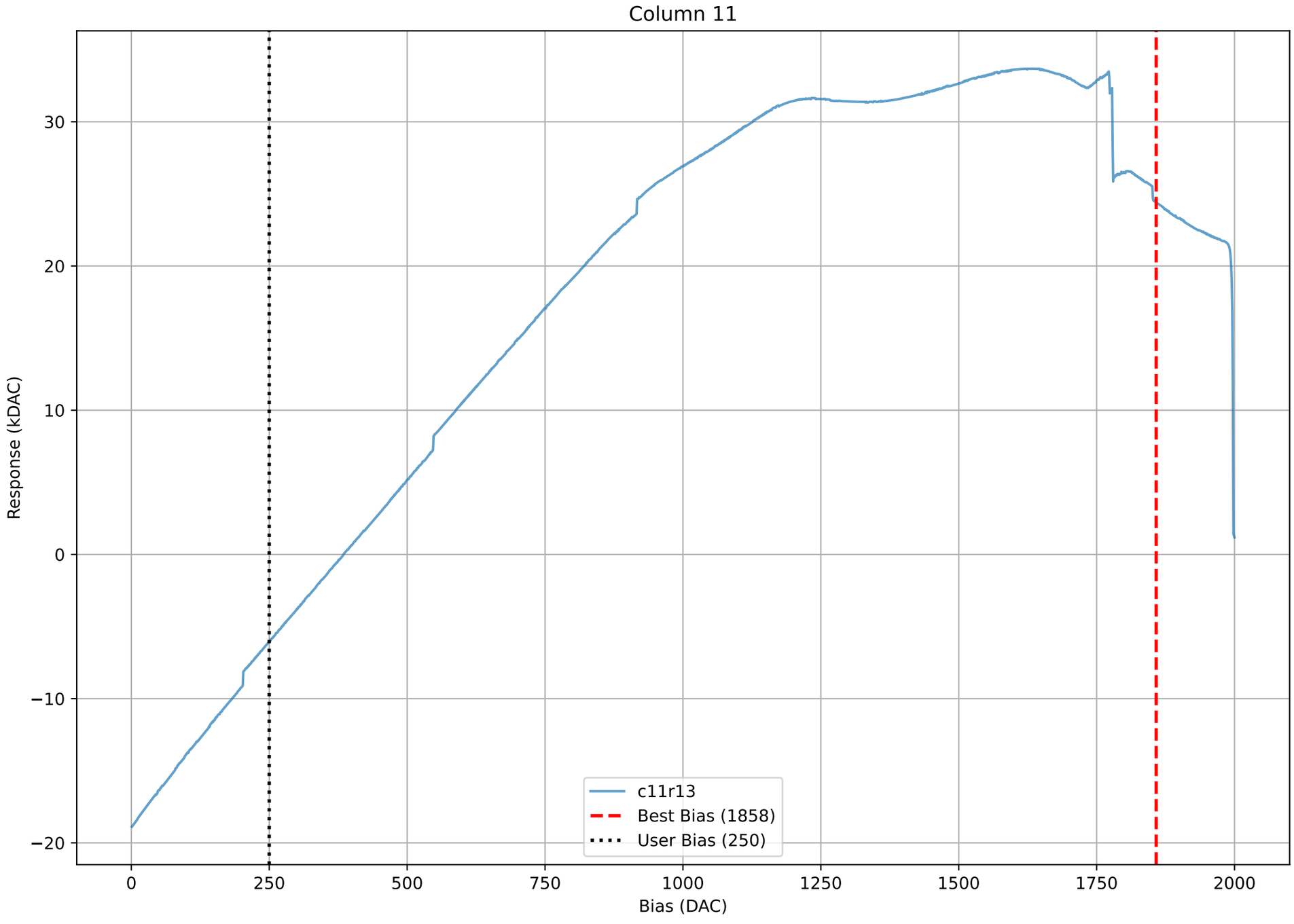

Column 6
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

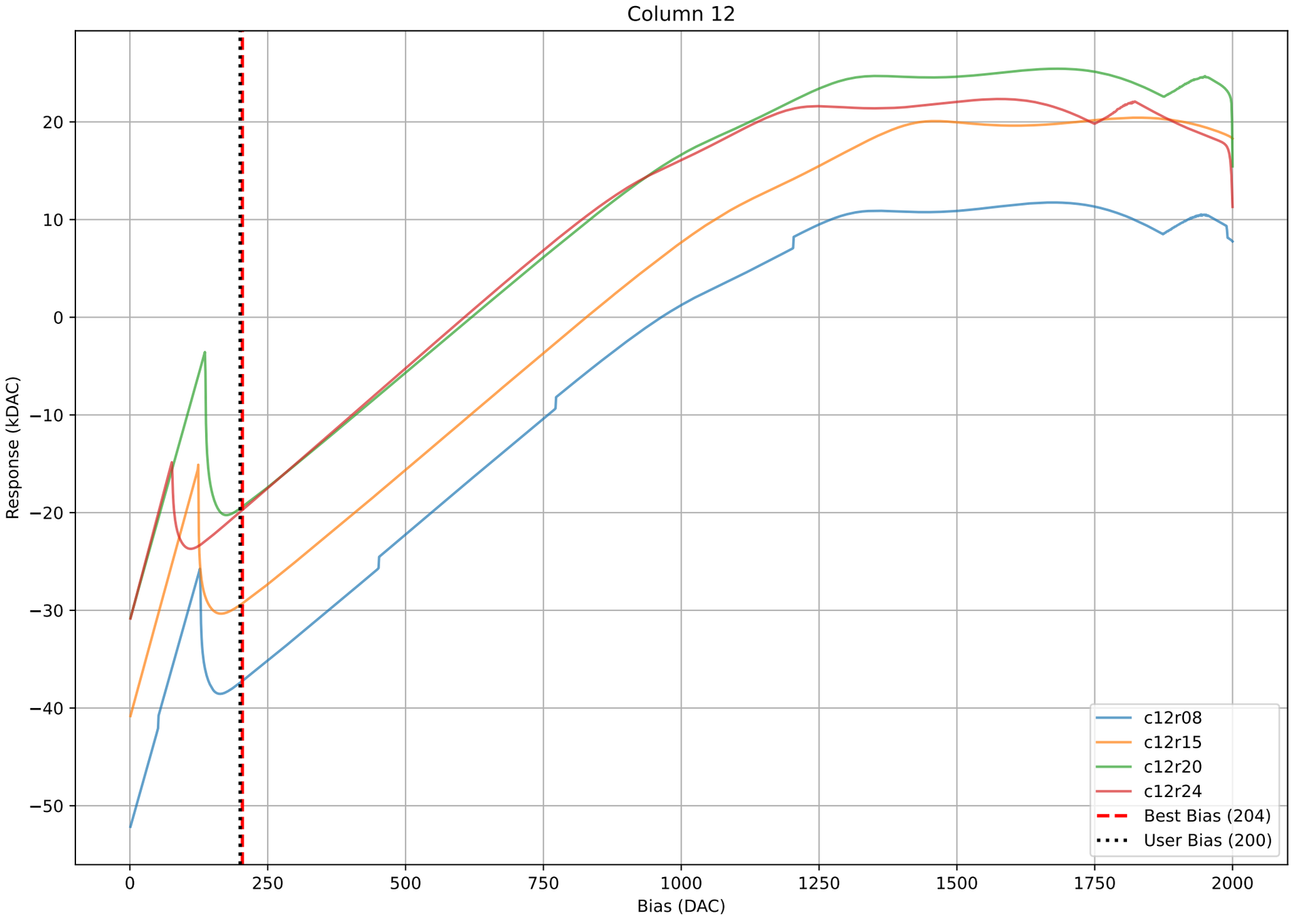

Column 7
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

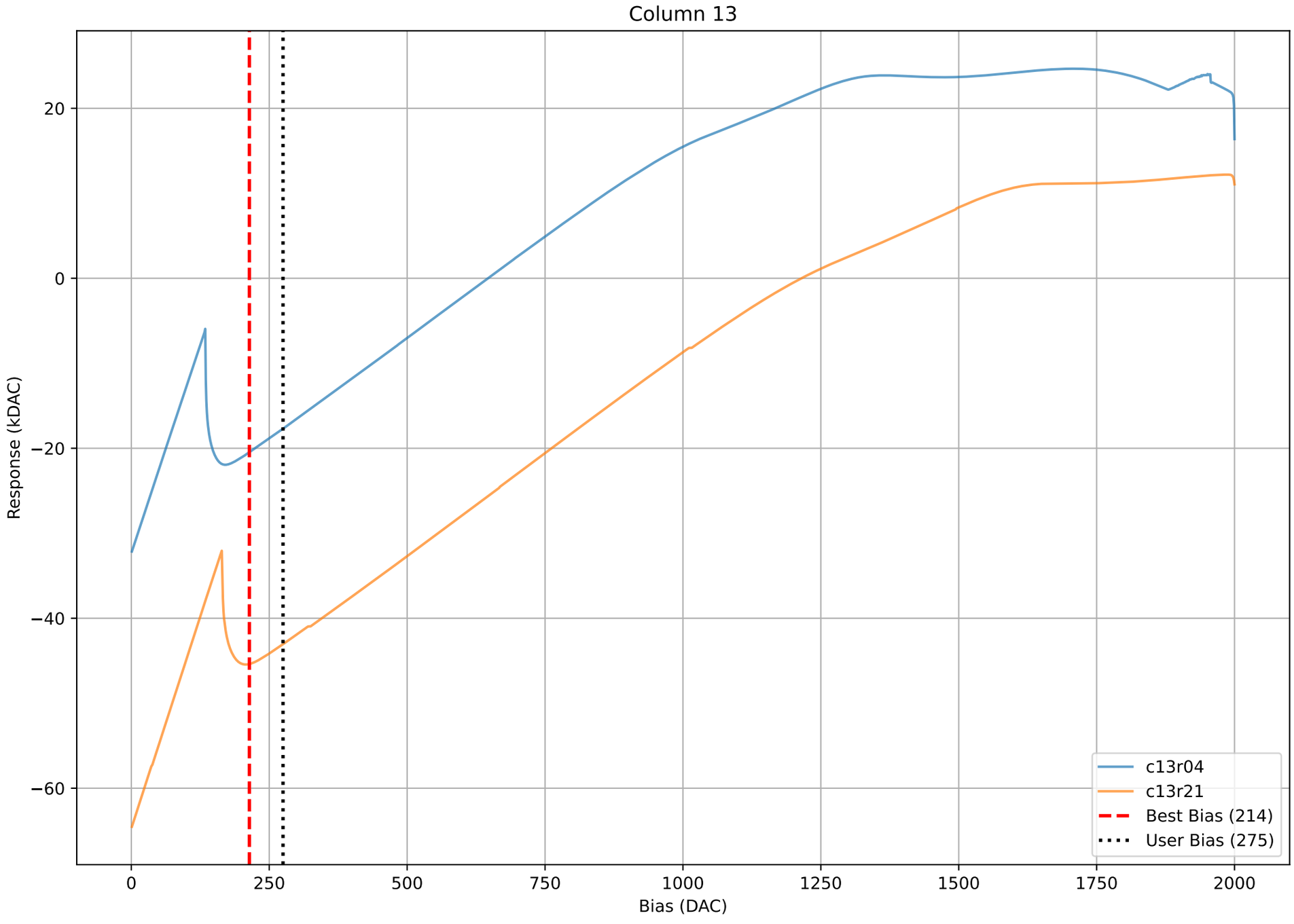

Column 8
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

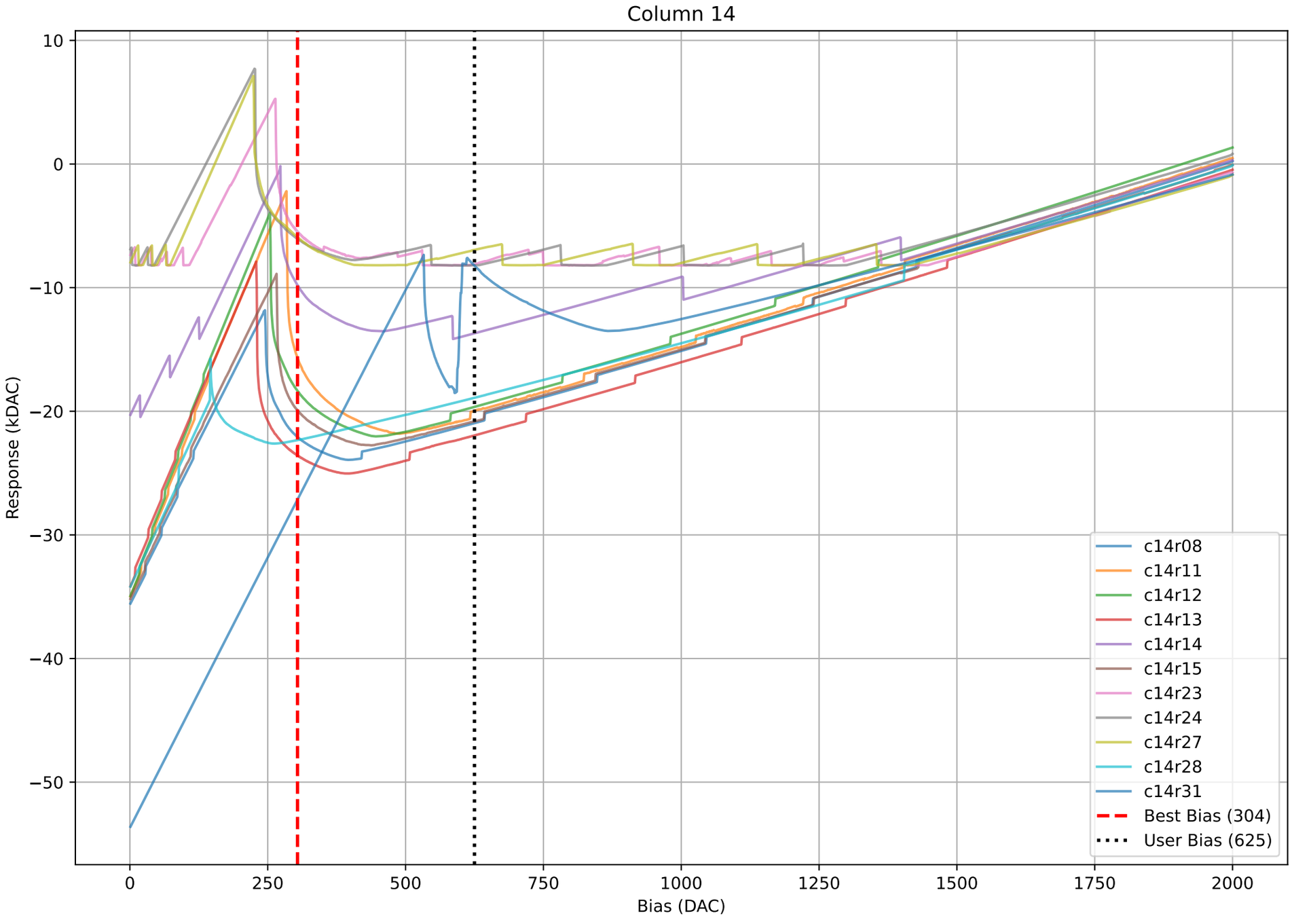

Column 9
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

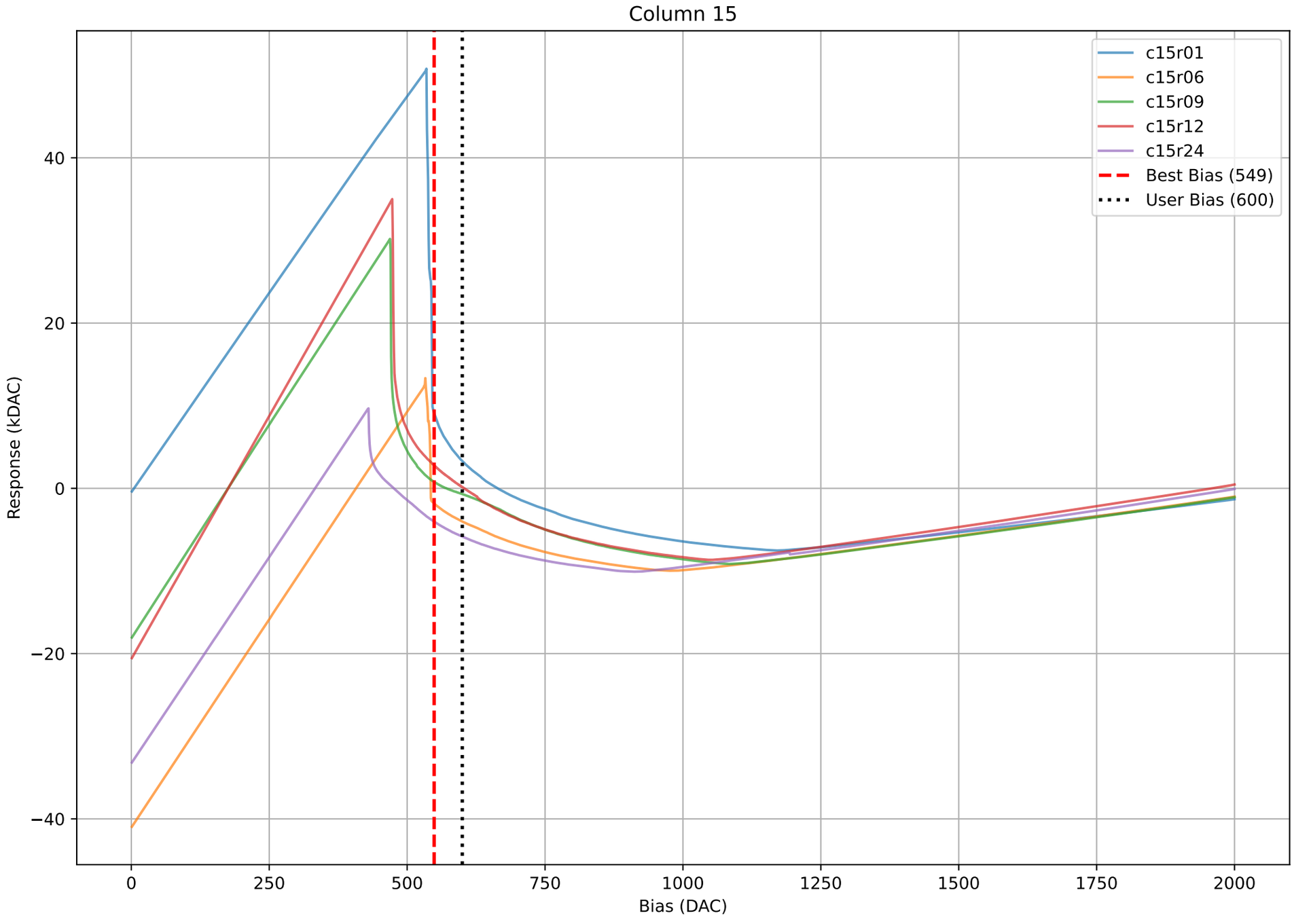

Column 10
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

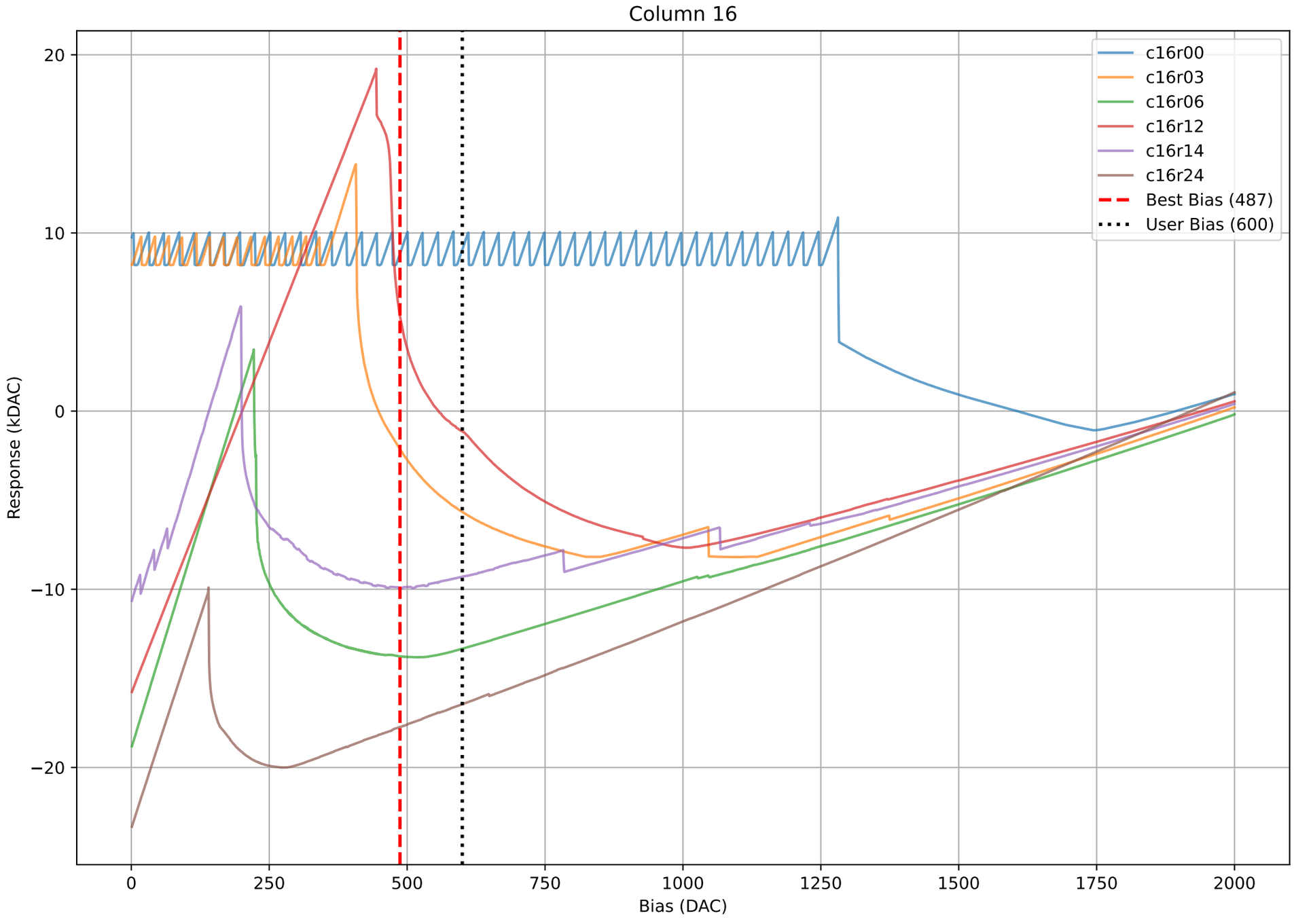

Column 11
Response (kDAC)
30
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c11r13
Best Bias (1858)
User Bias (250)

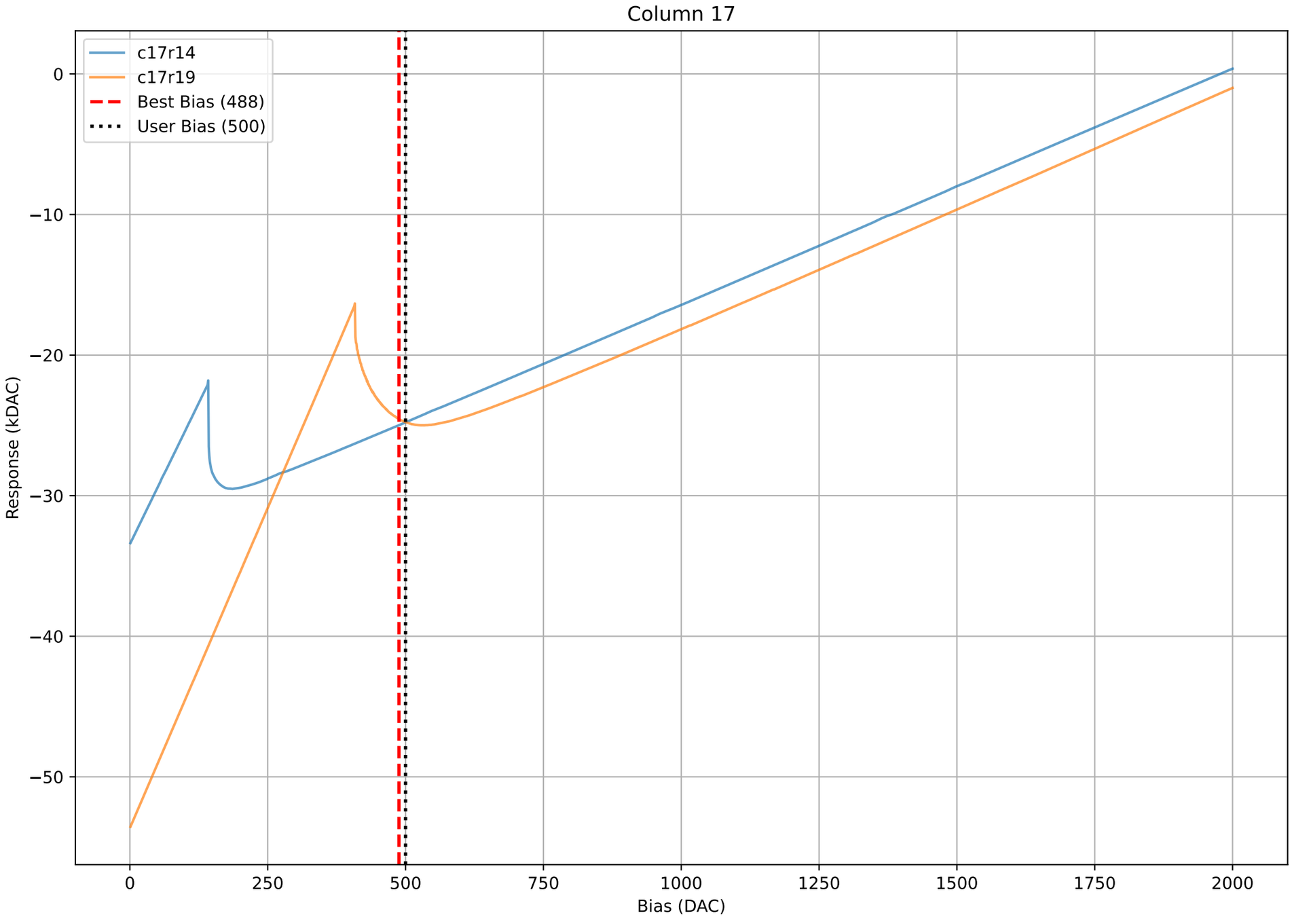

Column 12
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c12r08
c12r15
c12r20
c12r24
Best Bias (204)
User Bias (200)

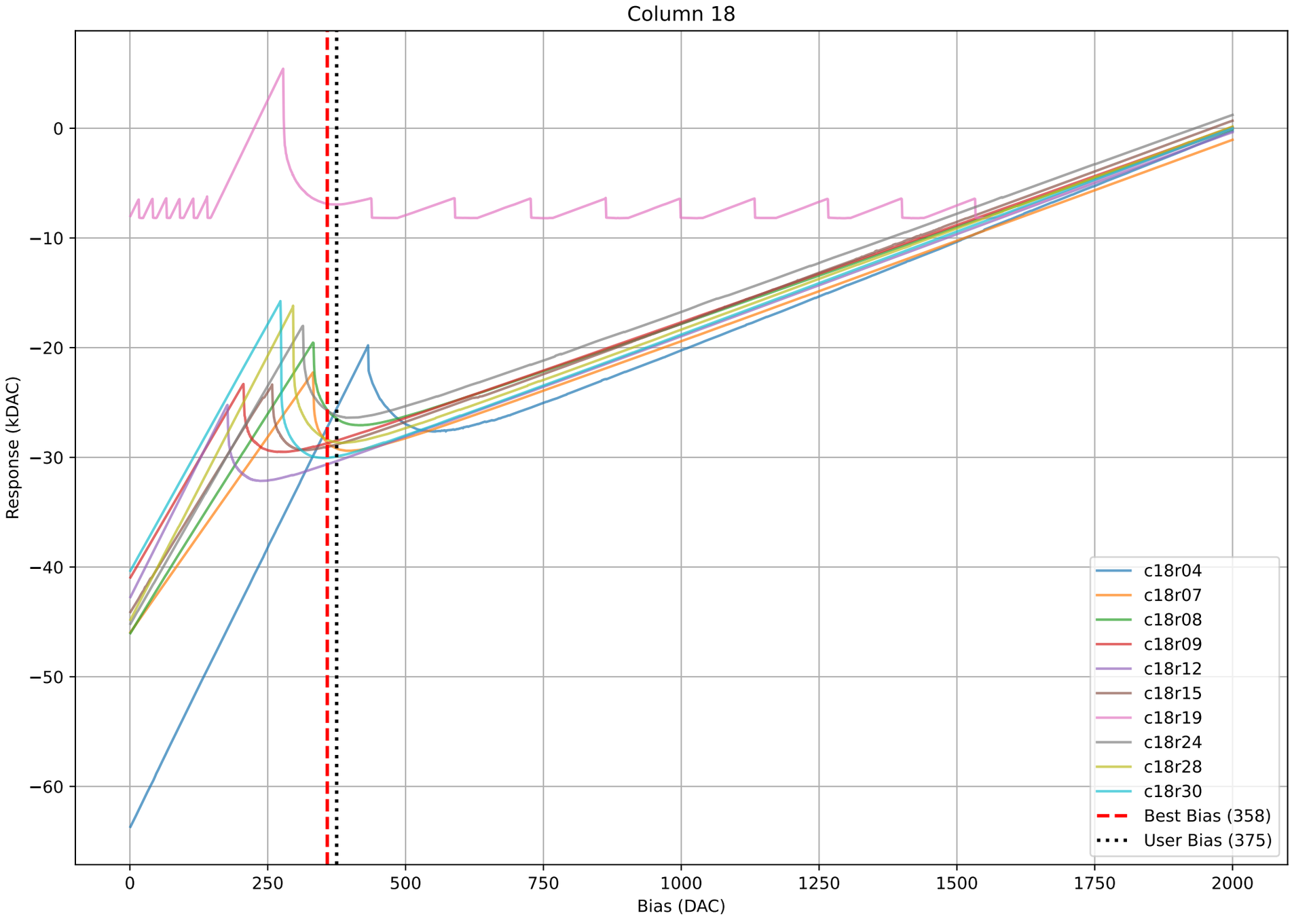

Column 13
Response (kDAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
c13r04
c13r21
Best Bias (214)
User Bias (275)

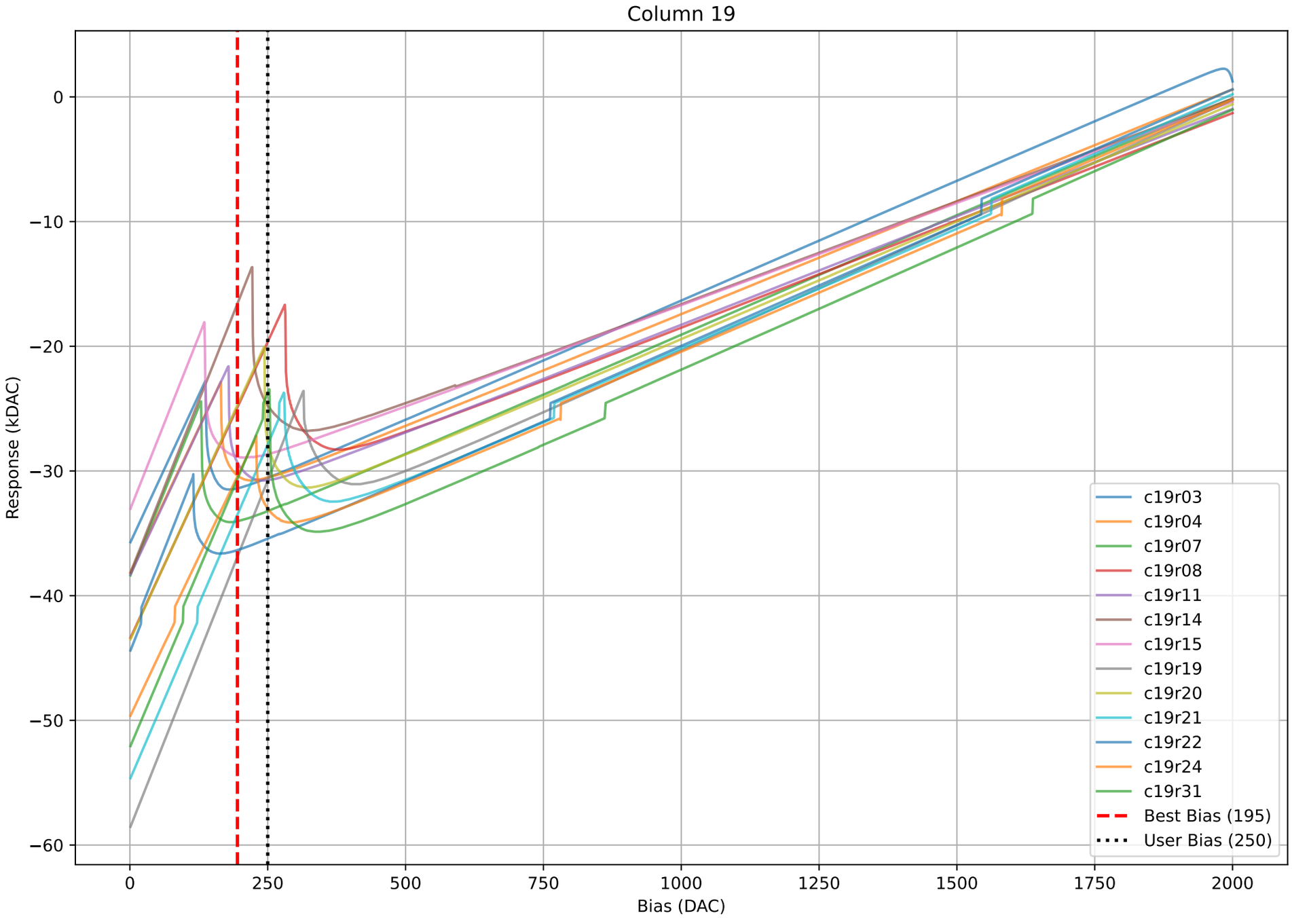

Column 14
Response (kDAC)
Bias (DAC)
10
0
−10
−20
−30
−40
−50
0
250
500
750
1000
1250
1500
1750
2000
c14r08
c14r11
c14r12
c14r13
c14r14
c14r15
c14r23
c14r24
c14r27
c14r28
c14r31
Best Bias (304)
User Bias (625)

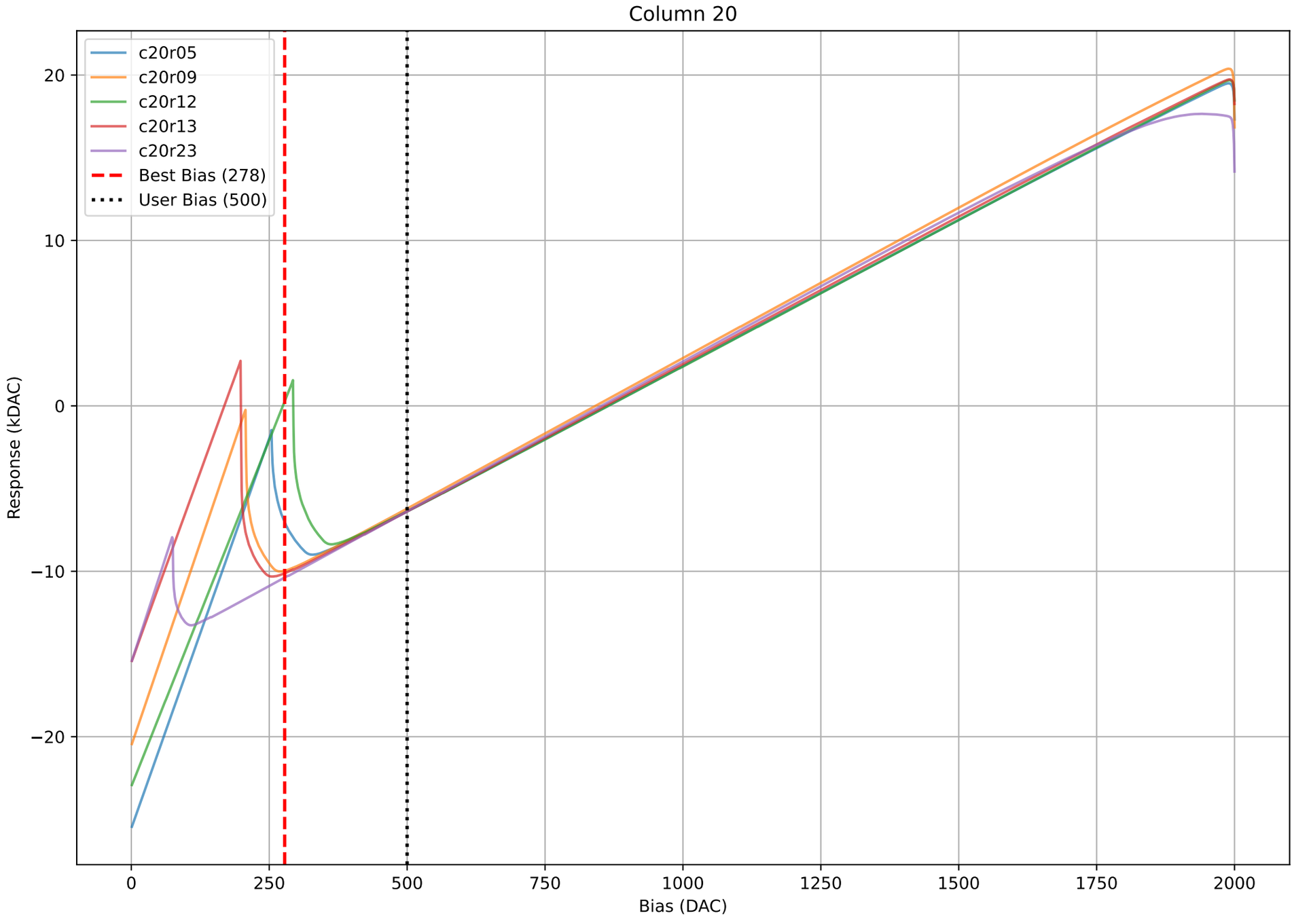

Column 15
c15r01
c15r06
c15r09
c15r12
c15r24
Best Bias (549)
User Bias (600)
Response (kDAC)
Bias (DAC)
40
20
0
−20
−40
0
250
500
750
1000
1250
1500
1750
2000

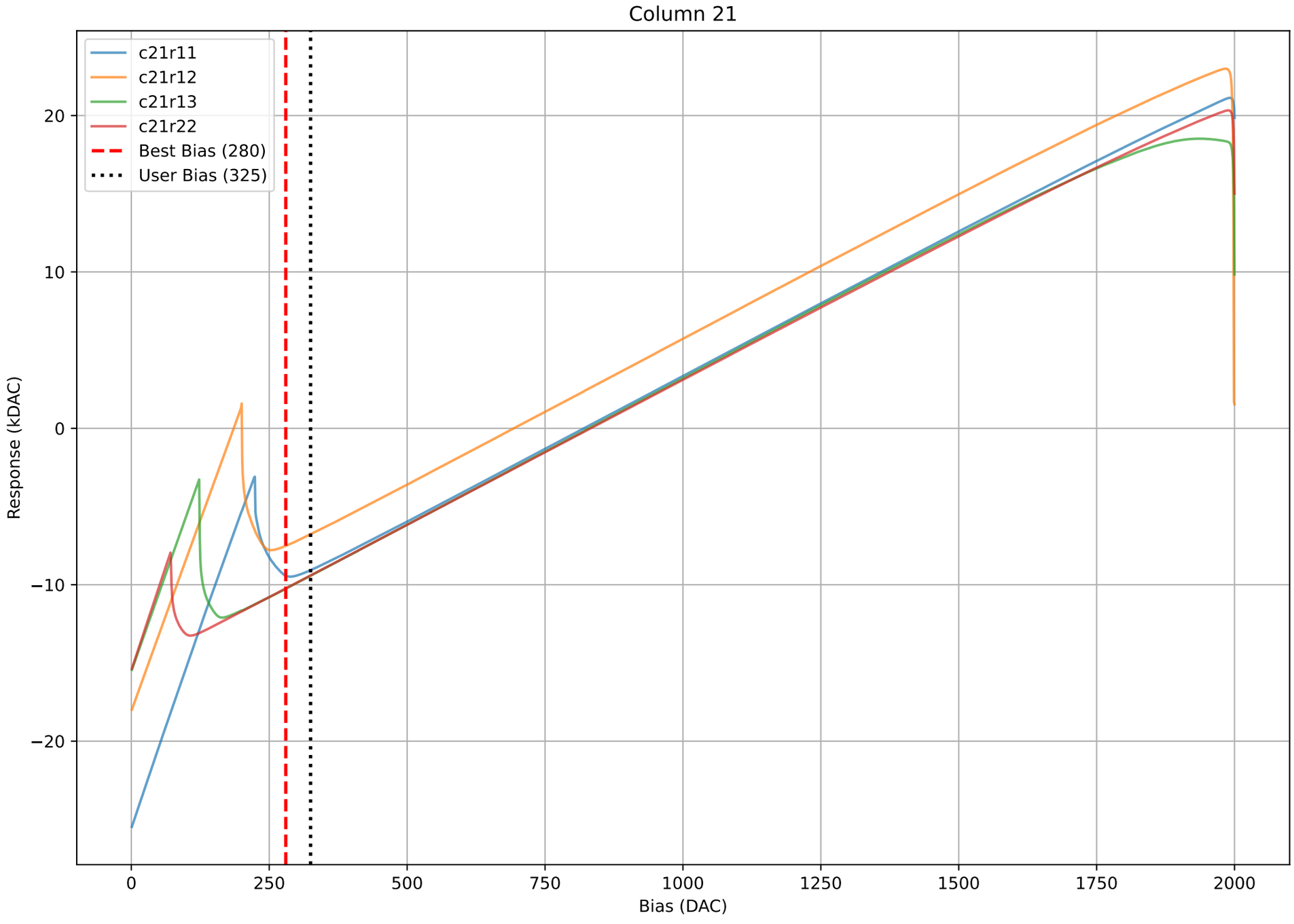

Column 16
c16r00
c16r03
c16r06
c16r12
c16r14
c16r24
Best Bias (487)
User Bias (600)
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

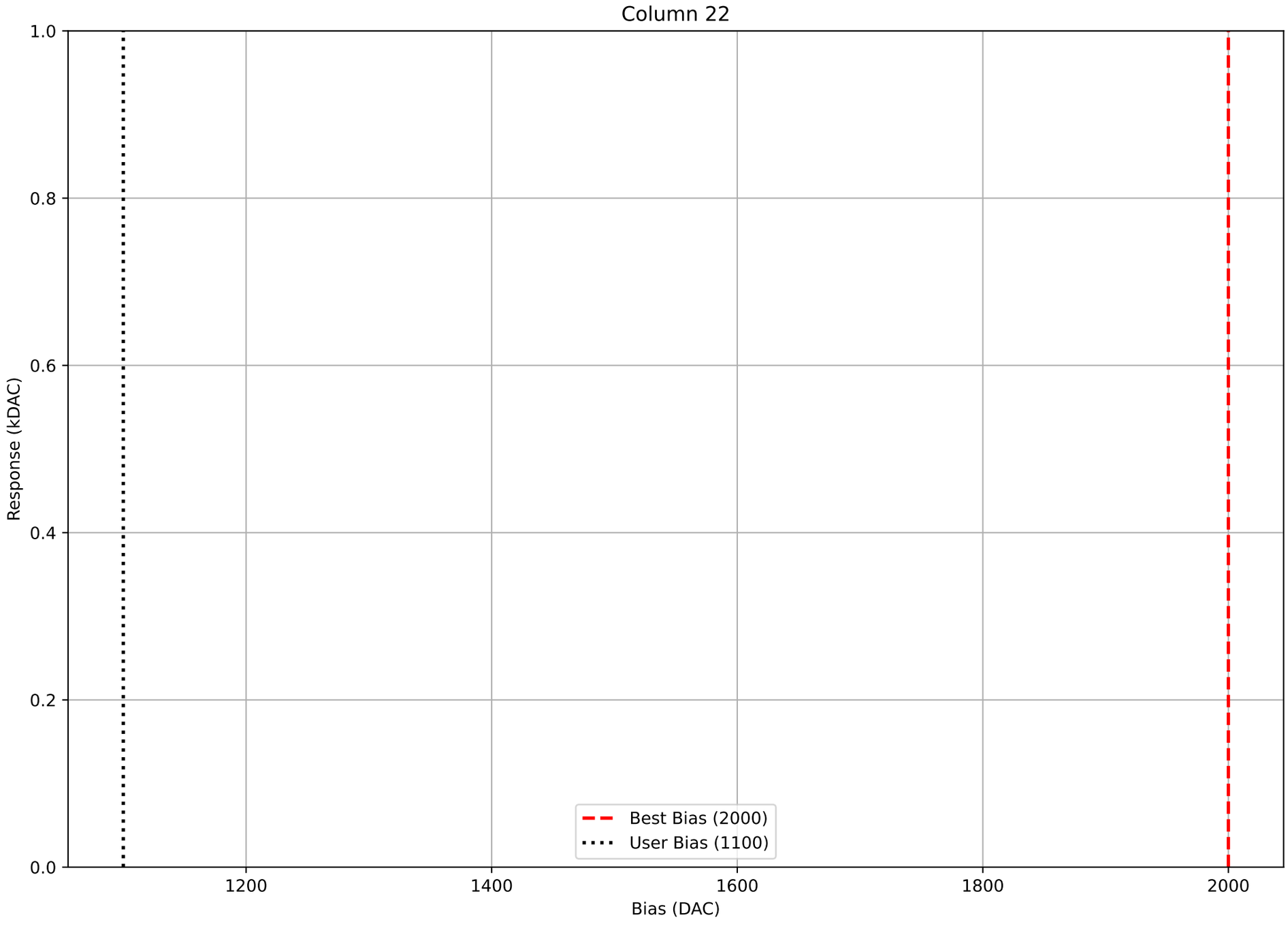

Column 17
c17r14
c17r19
Best Bias (488)
User Bias (500)
0
−10
−20
−30
−40
−50
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

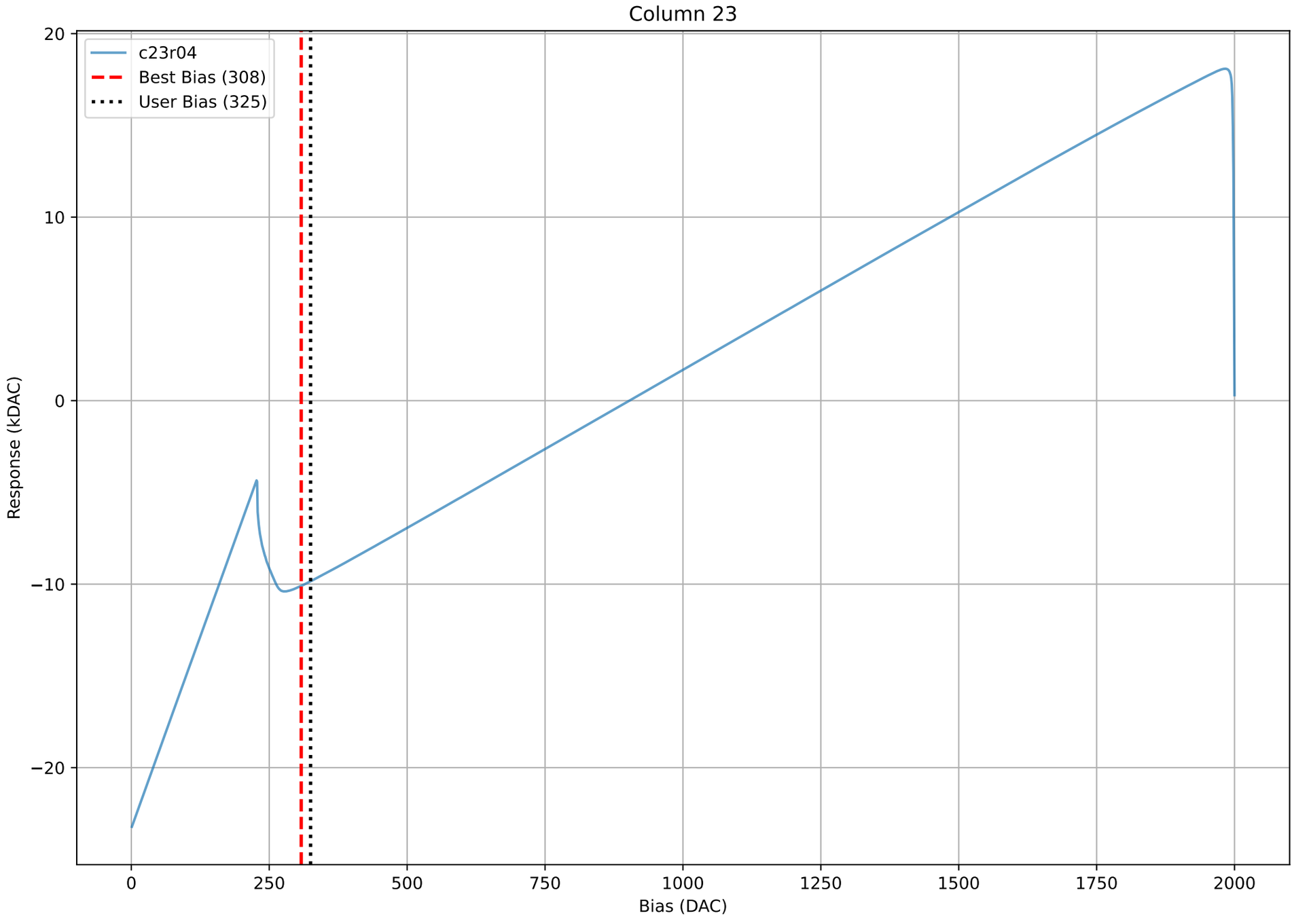

Column 18
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
−60
0
250
500
750
1000
1250
1500
1750
2000
c18r04
c18r07
c18r08
c18r09
c18r12
c18r15
c18r19
c18r24
c18r28
c18r30
Best Bias (358)
User Bias (375)

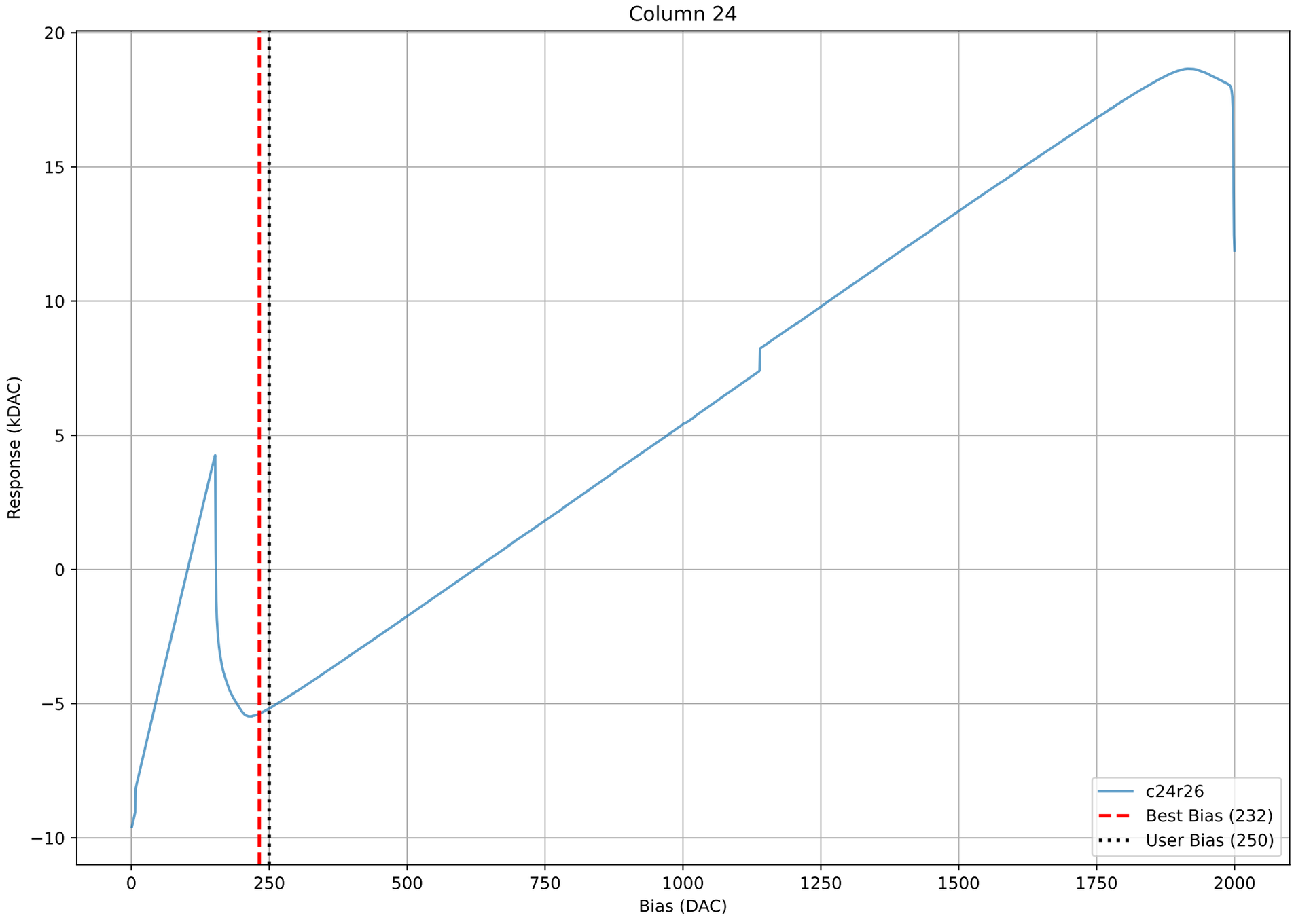

Column 19
Response (kDAC)
Bias (DAC)
0
−10
−20
−30
−40
−50
−60
0
250
500
750
1000
1250
1500
1750
2000
c19r03
c19r04
c19r07
c19r08
c19r11
c19r14
c19r15
c19r19
c19r20
c19r21
c19r22
c19r24
c19r31
Best Bias (195)
User Bias (250)

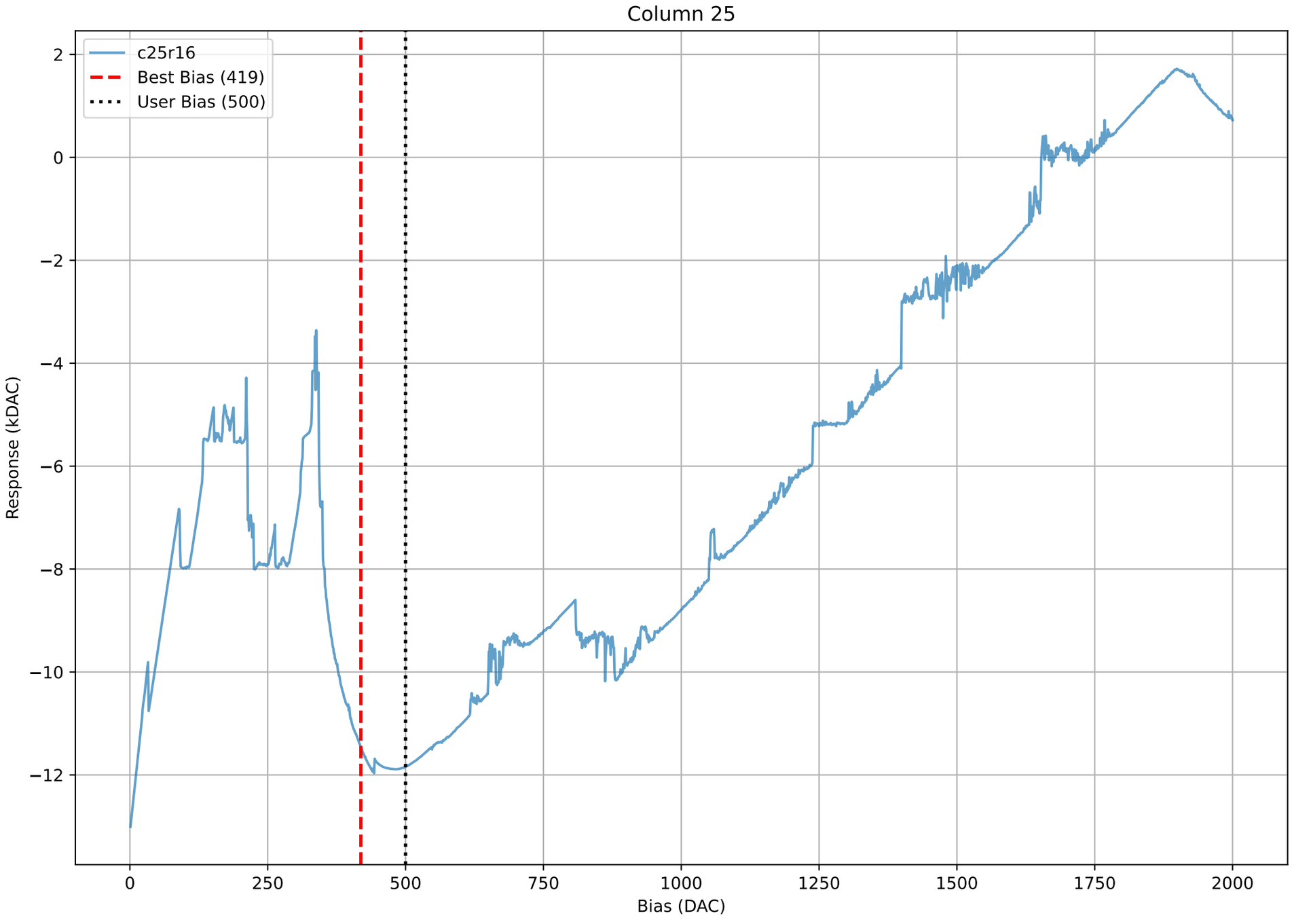

Column 20
c20r05
c20r09
c20r12
c20r13
c20r23
Best Bias (278)
User Bias (500)
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

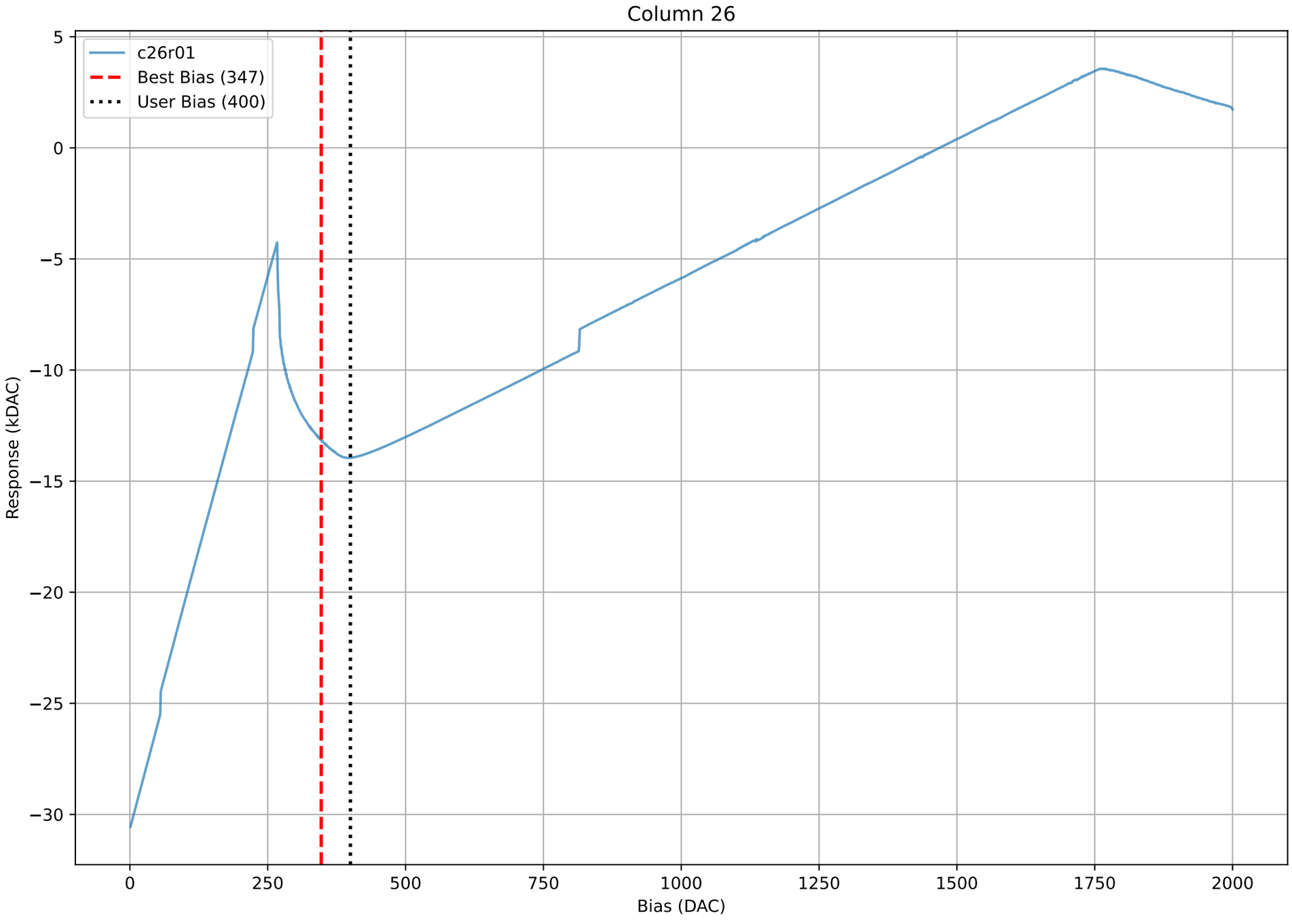

Column 21
c21r11
c21r12
c21r13
c21r22
Best Bias (280)
User Bias (325)
Response (kDAC)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)

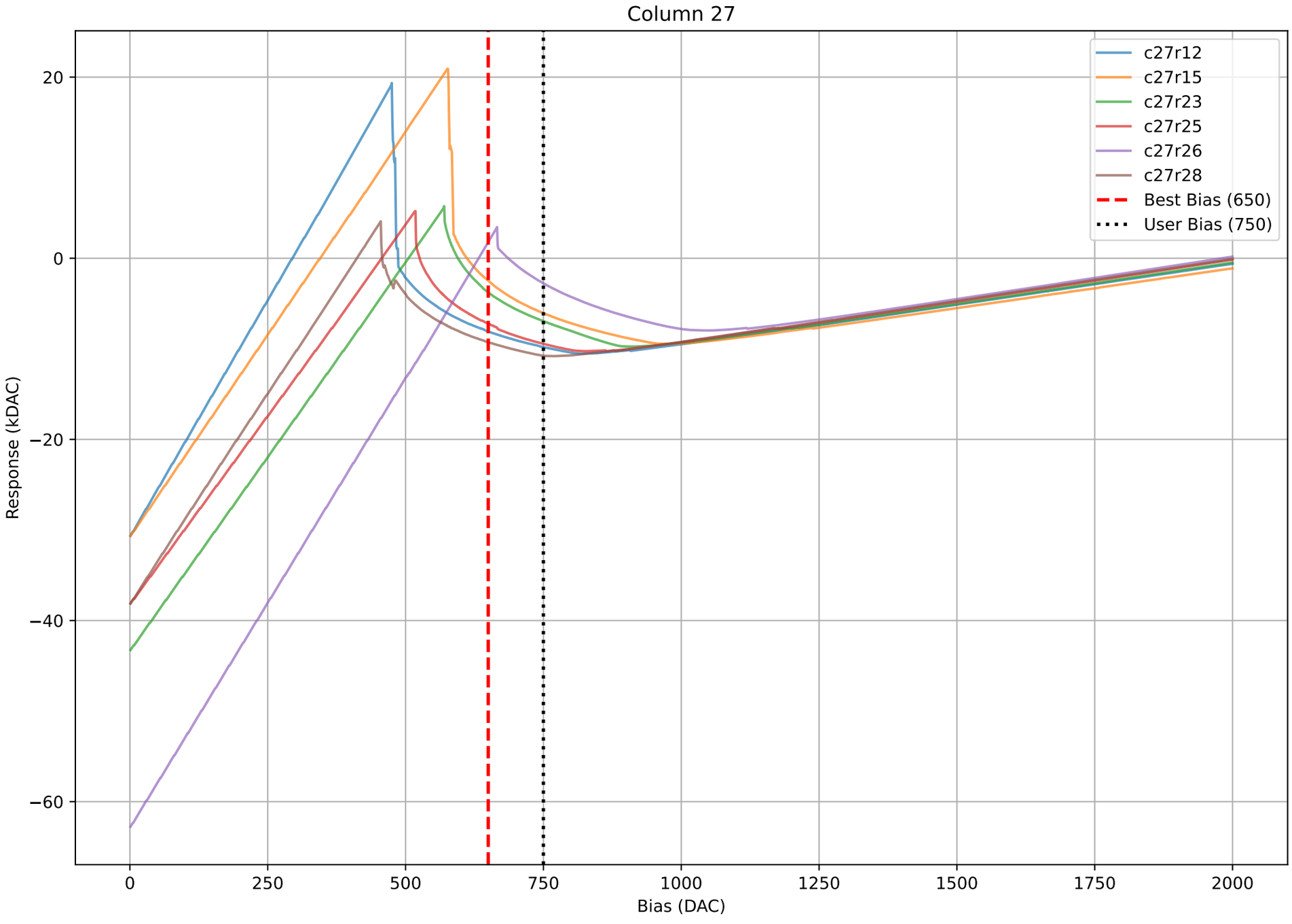

Column 22
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
1200
1400
1600
1800
2000
Bias (DAC)
Best Bias (2000)
User Bias (1100)

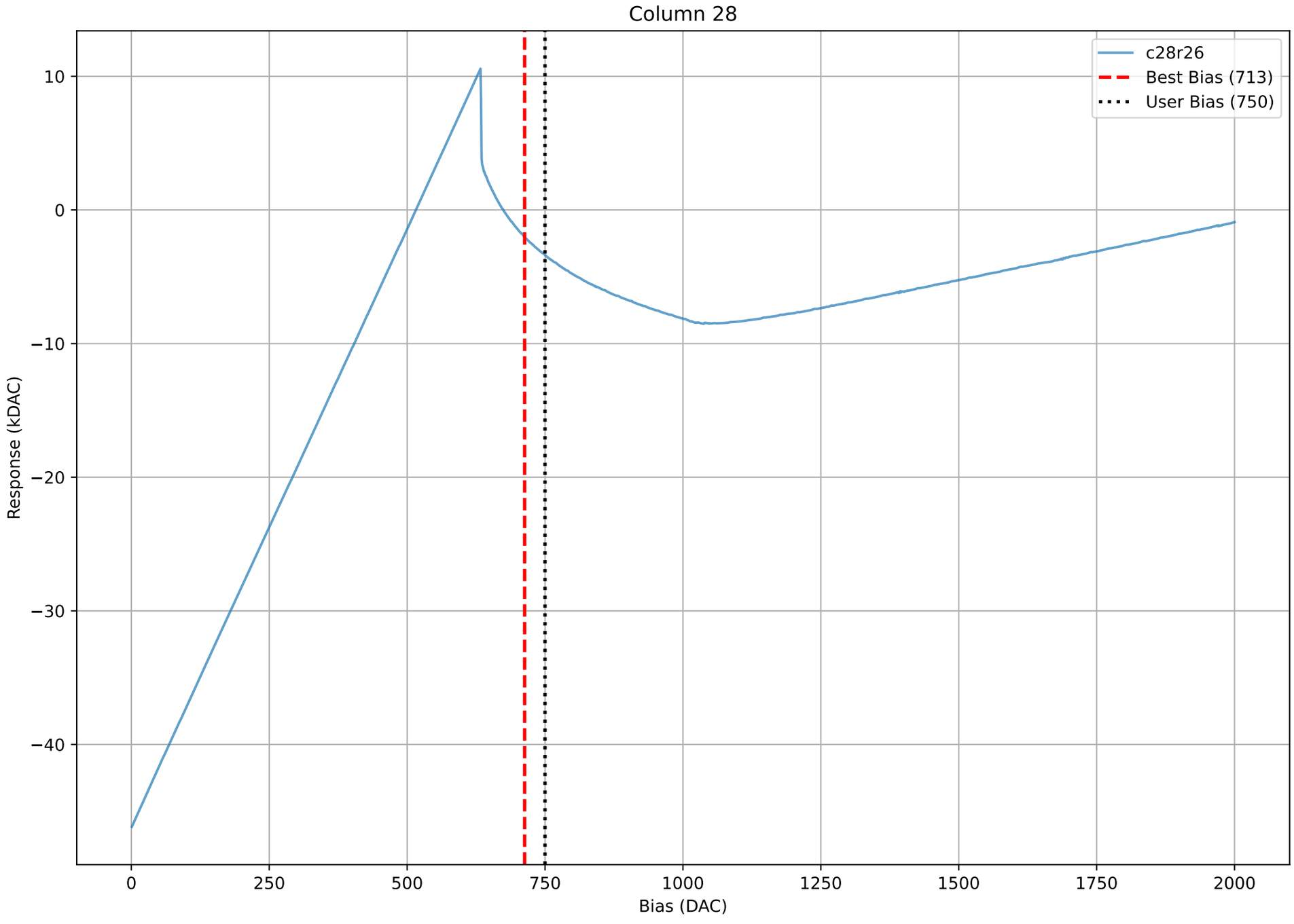

Column 23
c23r04
Best Bias (308)
User Bias (325)
Response (kDAC)
Bias (DAC)
20
10
0
−10
−20
0
250
500
750
1000
1250
1500
1750
2000

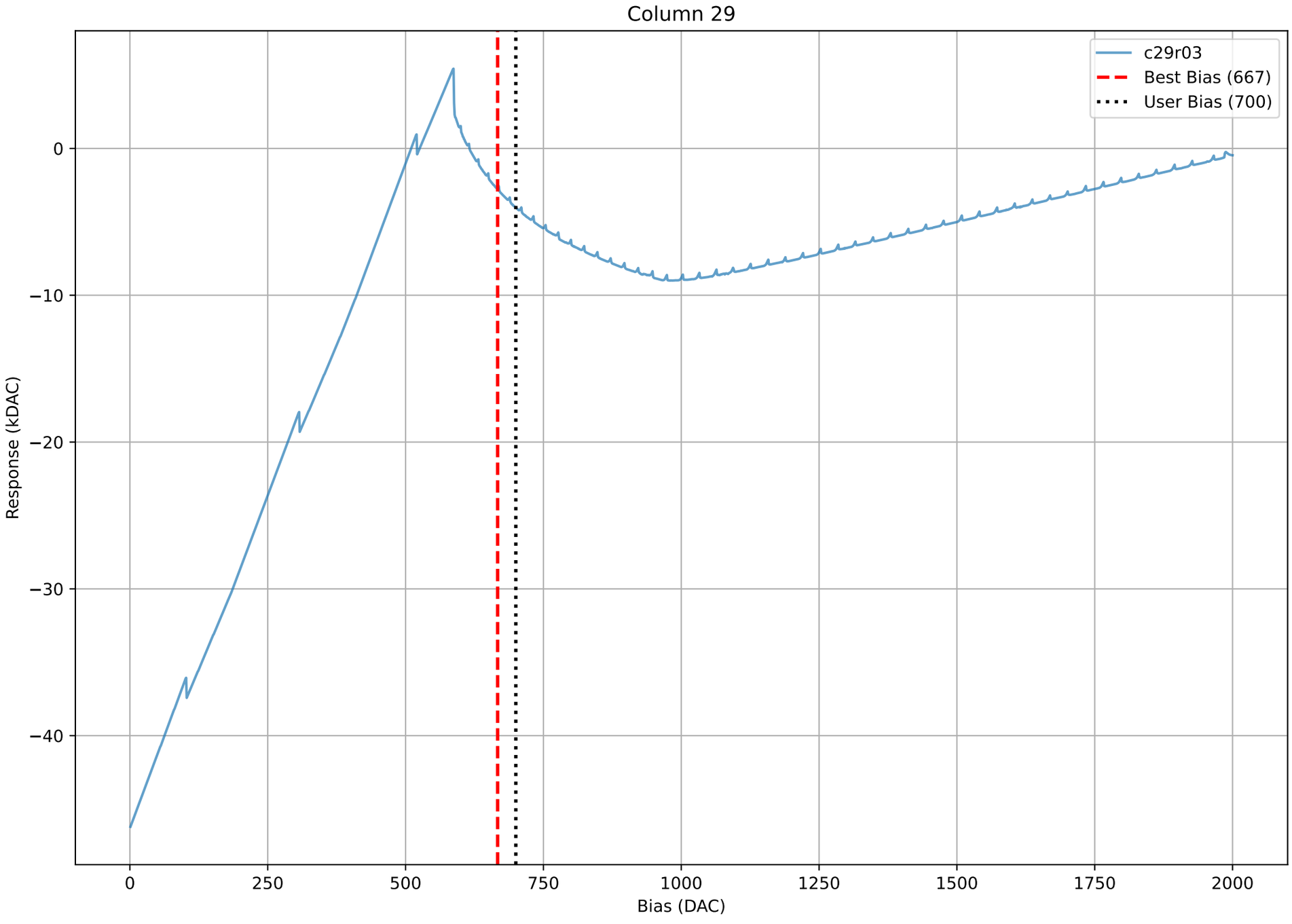

Column 24
Response (kDAC)
Bias (DAC)
20
15
10
5
0
−5
−10
0
250
500
750
1000
1250
1500
1750
2000
c24r26
Best Bias (232)
User Bias (250)

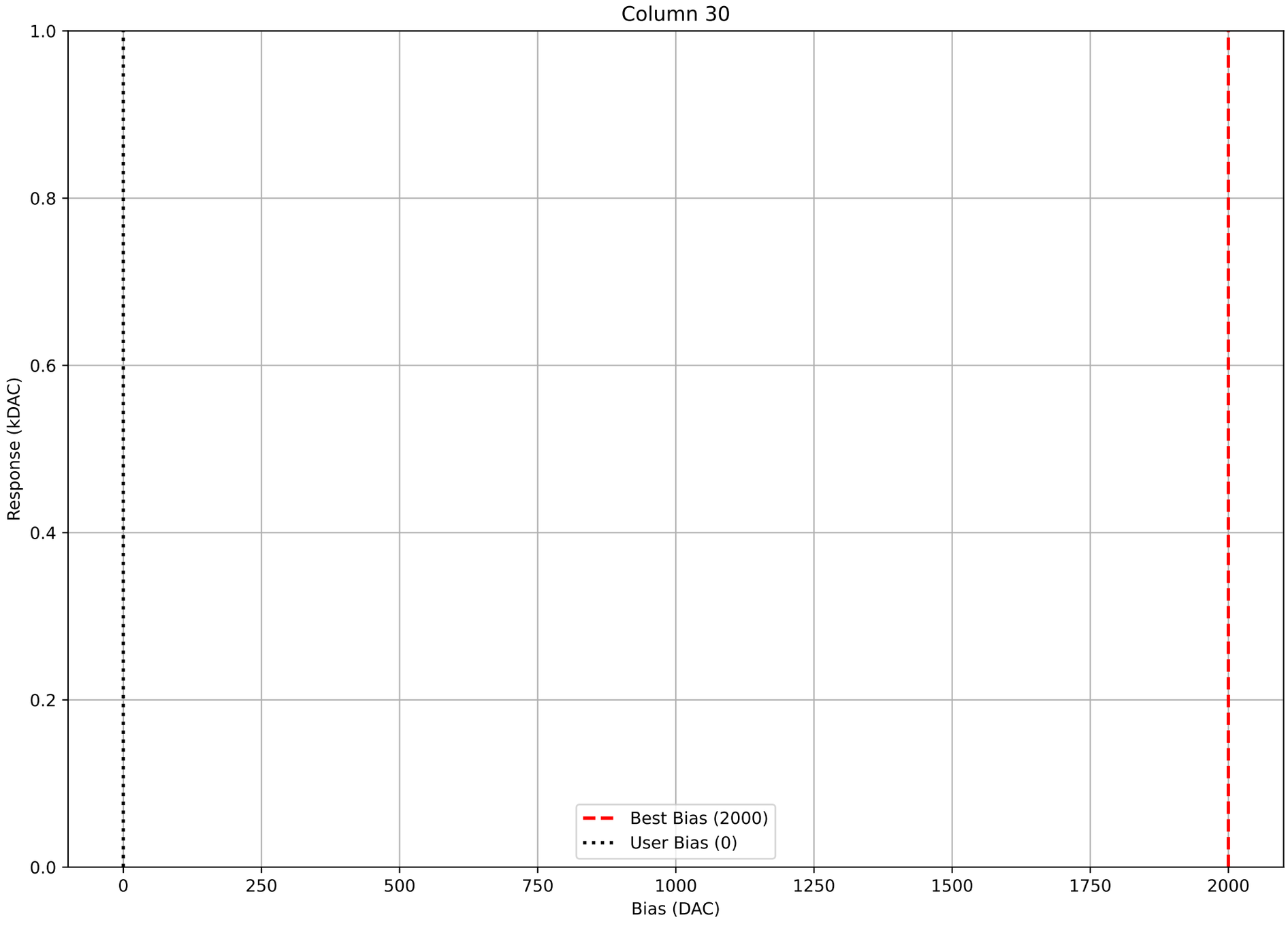

Column 25
c25r16
Best Bias (419)
User Bias (500)
Response (kDAC)
Bias (DAC)
2
0
−2
−4
−6
−8
−10
−12
0
250
500
750
1000
1250
1500
1750
2000

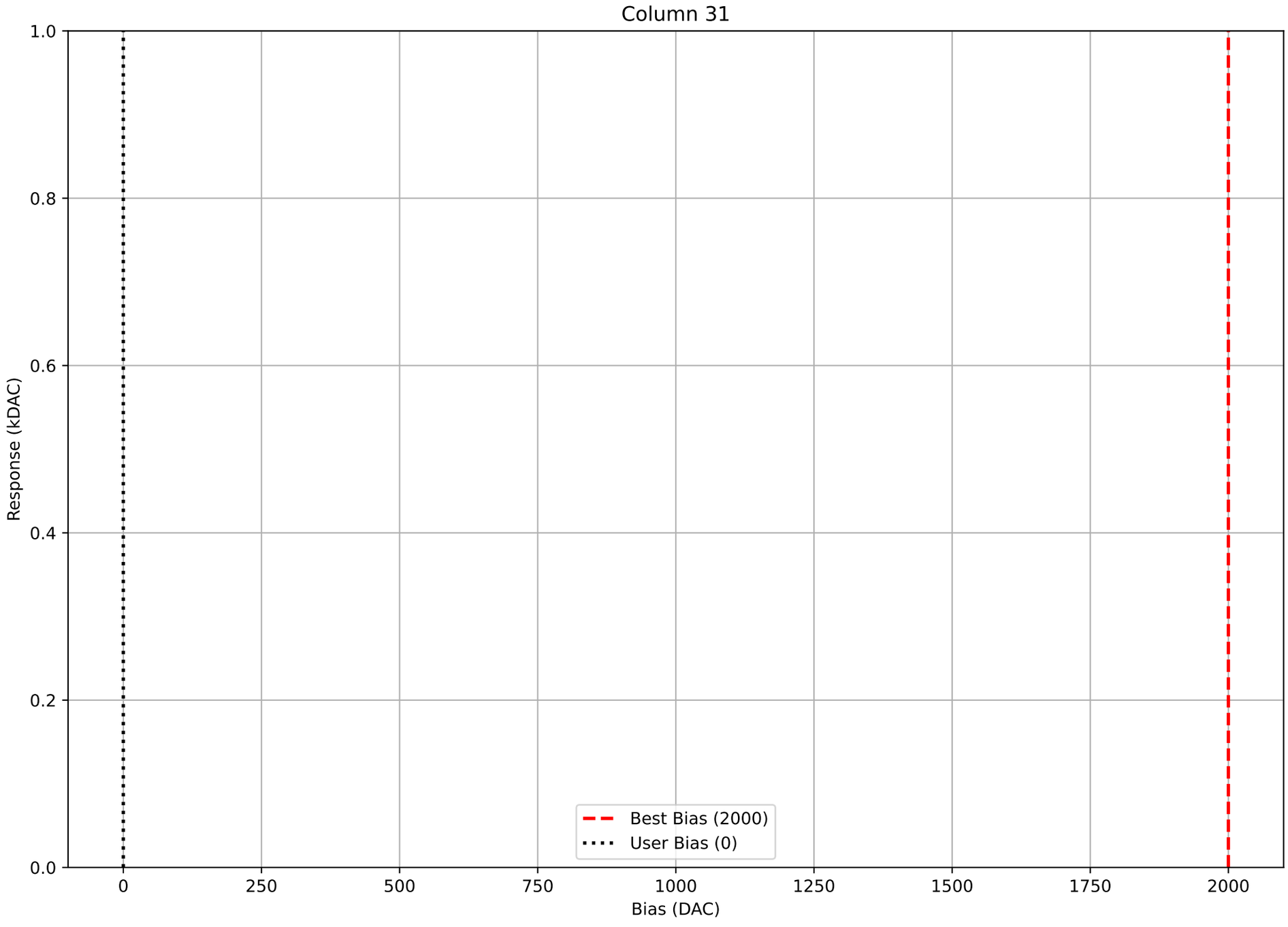

Column 26
c26r01
Best Bias (347)
User Bias (400)
Response (kDAC)
Bias (DAC)
5
0
−5
−10
−15
−20
−25
−30
0
250
500
750
1000
1250
1500
1750
2000


Column 27
c27r12
c27r15
c27r23
c27r25
c27r26
c27r28
Best Bias (650)
User Bias (750)
Response (kDAC)
Bias (DAC)
20
0
−20
−40
−60
0
250
500
750
1000
1250
1500
1750
2000


Column 28
c28r26
Best Bias (713)
User Bias (750)
Response (kDAC)
10
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)


Column 29
c29r03
Best Bias (667)
User Bias (700)
Response (kDAC)
0
−10
−20
−30
−40
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)


Column 30
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)


Column 31
1.0
0.8
0.6
0.4
0.2
0.0
Response (kDAC)
0
250
500
750
1000
1250
1500
1750
2000
Bias (DAC)
Best Bias (2000)
User Bias (0)

```
Column 0
  Noisy detectors excluded: []
  Best bias = 550.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [470.0, 550.0]: [(1, 0), (16, 0)]
    row=1 col=0 transition=[490.0, 470.0] min_slope=-2.697 mean_slope=-0.509 drop_kdac=-10.698
    row=16 col=0 transition=[482.0, 472.0] min_slope=-1.173 mean_slope=-0.430 drop_kdac=-4.705
  User bias = 550
  Clean detectors in transition in [470.0, 550.0]: [(1, 0), (16, 0)]
    row=1 col=0 transition=[490.0, 470.0] min_slope=-2.697 mean_slope=-0.509 drop_kdac=-10.698
    row=16 col=0 transition=[482.0, 472.0] min_slope=-1.173 mean_slope=-0.430 drop_kdac=-4.705
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 1
  Noisy detectors excluded: []
  Best bias = 460.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [380.0, 460.0]: [(21, 1), (22, 1)]
    row=21 col=1 transition=[403.0, 380.0] min_slope=-3.522 mean_slope=-0.518 drop_kdac=-12.534
    row=22 col=1 transition=[443.0, 423.0] min_slope=-6.019 mean_slope=-0.822 drop_kdac=-17.352
  User bias = 600
  Clean detectors in transition in [520.0, 600.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=26 col=1 transition=[532.0, 509.0] min_slope=-8.343 mean_slope=-0.940 drop_kdac=-22.569
gap_to_window=11.0 reason=accepted transition outside user window

Column 2
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 600
  Clean detectors in transition in [520.0, 600.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 3
  Noisy detectors excluded: [(18, 3)]
    row=18 col=3 noisy_transition=[1477.0, 1473.0] min_slope=-1.181 mean_slope=-0.576 drop_kdac=-2.968
    row=18 col=3 noisy_transition=[1436.0, 1426.0] min_slope=-1.891 mean_slope=-0.811 drop_kdac=-8.913
    row=18 col=3 noisy_transition=[1409.0, 1405.0] min_slope=-1.081 mean_slope=-0.460 drop_kdac=-2.303
    row=18 col=3 noisy_transition=[521.0, 515.0] min_slope=-3.204 mean_slope=-1.566 drop_kdac=-10.870
    row=18 col=3 noisy_transition=[476.0, 474.0] min_slope=-1.433 mean_slope=-0.945 drop_kdac=-2.866
```

```
    row=18 col=3 noisy_transition=[339.0, 326.0] min_slope=-2.238 mean_slope=-0.897 drop_kdac=-12.637
    row=18 col=3 noisy_transition=[114.0, 112.0] min_slope=-1.840 mean_slope=-1.237 drop_kdac=-3.681
    row=18 col=3 noisy_transition=[97.0, 96.0] min_slope=-1.269 mean_slope=-1.257 drop_kdac=-2.416
  Best bias = 189.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [109.0, 189.0]: [(19, 3), (20, 3), (23, 3)]
    row=19 col=3 transition=[168.0, 150.0] min_slope=-2.251 mean_slope=-0.457 drop_kdac=-8.700
    row=20 col=3 transition=[122.0, 109.0] min_slope=-1.100 mean_slope=-0.324 drop_kdac=-4.394
    row=23 col=3 transition=[154.0, 141.0] min_slope=-1.888 mean_slope=-0.466 drop_kdac=-6.533
  User bias = 375
  Clean detectors in transition in [295.0, 375.0]: [(1, 3)]
    row=1 col=3 transition=[338.0, 317.0] min_slope=-3.059 mean_slope=-0.529 drop_kdac=-11.647
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=11 col=3 transition=[376.0, 351.0] min_slope=-3.828 mean_slope=-0.574 drop_kdac=-14.935
gap_to_window=1.0 reason=accepted transition outside user window
    row=12 col=3 transition=[275.0, 258.0] min_slope=-3.040 mean_slope=-0.559 drop_kdac=-10.041
gap_to_window=37.0 reason=accepted transition outside user window
    row=18 col=3 transition=[339.0, 326.0] min_slope=-2.238 mean_slope=-0.897 drop_kdac=-12.637
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)

Column 4
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 5
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 6
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
```

```
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 7
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 8
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 9
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 10
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []
```

```
Column 11
  Noisy detectors excluded: [(0, 11)]
    row=0 col=11 noisy_transition=[1552.0, 1544.0] min_slope=-1.786 mean_slope=-0.818 drop_kdac=-7.523
    row=0 col=11 noisy_transition=[1531.0, 1527.0] min_slope=-1.429 mean_slope=-0.664 drop_kdac=-3.215
    row=0 col=11 noisy_transition=[1515.0, 1514.0] min_slope=-1.955 mean_slope=-1.534 drop_kdac=-2.569
    row=0 col=11 noisy_transition=[1482.0, 1473.0] min_slope=-1.565 mean_slope=-0.422 drop_kdac=-4.045
    row=0 col=11 noisy_transition=[1289.0, 1288.0] min_slope=-1.445 mean_slope=-1.434 drop_kdac=-2.876
    row=0 col=11 noisy_transition=[1277.0, 1275.0] min_slope=-1.665 mean_slope=-1.101 drop_kdac=-3.331
    row=0 col=11 noisy_transition=[336.0, 333.0] min_slope=-1.037 mean_slope=-0.541 drop_kdac=-2.163
  Best bias = 1858.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [1778.0, 1858.0]: [(13, 11)]
    row=13 col=11 transition=[1780.0, 1778.0] min_slope=-3.241 mean_slope=-2.053 drop_kdac=-6.482
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=16 col=11 transition=[256.0, 255.0] min_slope=-0.096 mean_slope=-0.089 drop_kdac=-0.228
gap_to_window=6.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=16 col=11 transition=[136.0, 135.0] min_slope=-0.108 mean_slope=-0.101 drop_kdac=-0.253
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 12
  Noisy detectors excluded: []
  Best bias = 204.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [124.0, 204.0]: [(8, 12), (15, 12), (20, 12)]
    row=8 col=12 transition=[151.0, 127.0] min_slope=-2.810 mean_slope=-0.494 drop_kdac=-12.411
    row=15 col=12 transition=[145.0, 124.0] min_slope=-4.535 mean_slope=-0.649 drop_kdac=-14.509
    row=20 col=12 transition=[161.0, 136.0] min_slope=-3.630 mean_slope=-0.619 drop_kdac=-16.166
  User bias = 200
  Clean detectors in transition in [120.0, 200.0]: [(8, 12), (15, 12), (20, 12)]
    row=8 col=12 transition=[151.0, 127.0] min_slope=-2.810 mean_slope=-0.494 drop_kdac=-12.411
    row=15 col=12 transition=[145.0, 124.0] min_slope=-4.535 mean_slope=-0.649 drop_kdac=-14.509
    row=20 col=12 transition=[161.0, 136.0] min_slope=-3.630 mean_slope=-0.619 drop_kdac=-16.166
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=7 col=12 transition=[115.0, 98.0] min_slope=-0.629 mean_slope=-0.231 drop_kdac=-4.216
gap_to_window=22.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=12 transition=[107.0, 90.0] min_slope=-0.785 mean_slope=-0.271 drop_kdac=-4.598
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 13
  Noisy detectors excluded: []
  Best bias = 214.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [134.0, 214.0]: [(4, 13), (21, 13)]
```

```
    row=4 col=13 transition=[157.0, 134.0] min_slope=-4.243 mean_slope=-0.641 drop_kdac=-15.502
    row=21 col=13 transition=[189.0, 164.0] min_slope=-2.807 mean_slope=-0.488 drop_kdac=-12.730
  User bias = 275
  Clean detectors in transition in [195.0, 275.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=7 col=13 transition=[267.0, 262.0] min_slope=-0.816 mean_slope=-0.371 drop_kdac=-2.162
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=21 col=13 transition=[189.0, 164.0] min_slope=-2.807 mean_slope=-0.488 drop_kdac=-12.730
gap_to_window=31.0 reason=accepted transition outside user window
    row=30 col=13 transition=[277.0, 276.0] min_slope=-0.239 mean_slope=-0.236 drop_kdac=-0.518
gap_to_window=2.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=30 col=13 transition=[157.0, 156.0] min_slope=-0.323 mean_slope=-0.317 drop_kdac=-0.678
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 14
  Noisy detectors excluded: [(22, 14), (29, 14)]
    row=22 col=14 noisy_transition=[1254.0, 1253.0] min_slope=-1.071 mean_slope=-1.023 drop_kdac=-2.142
    row=22 col=14 noisy_transition=[1061.0, 1060.0] min_slope=-1.045 mean_slope=-1.035 drop_kdac=-2.090
    row=22 col=14 noisy_transition=[862.0, 861.0] min_slope=-1.138 mean_slope=-1.095 drop_kdac=-2.276
    row=22 col=14 noisy_transition=[654.0, 653.0] min_slope=-1.143 mean_slope=-1.091 drop_kdac=-2.286
    row=22 col=14 noisy_transition=[439.0, 438.0] min_slope=-1.052 mean_slope=-1.041 drop_kdac=-2.058
    row=22 col=14 noisy_transition=[203.0, 182.0] min_slope=-3.034 mean_slope=-0.417 drop_kdac=-9.303
    row=29 col=14 noisy_transition=[1402.0, 1401.0] min_slope=-1.124 mean_slope=-1.119 drop_kdac=-2.226
    row=29 col=14 noisy_transition=[1201.0, 1200.0] min_slope=-1.116 mean_slope=-1.114 drop_kdac=-2.225
    row=29 col=14 noisy_transition=[992.0, 991.0] min_slope=-1.088 mean_slope=-1.087 drop_kdac=-2.174
    row=29 col=14 noisy_transition=[776.0, 775.0] min_slope=-1.083 mean_slope=-1.075 drop_kdac=-2.135
    row=29 col=14 noisy_transition=[540.0, 539.0] min_slope=-1.037 mean_slope=-1.036 drop_kdac=-2.072
    row=29 col=14 noisy_transition=[270.0, 241.0] min_slope=-2.451 mean_slope=-0.321 drop_kdac=-9.641
    row=29 col=14 noisy_transition=[88.0, 87.0] min_slope=-1.072 mean_slope=-1.063 drop_kdac=-2.139
    row=29 col=14 noisy_transition=[62.0, 61.0] min_slope=-1.059 mean_slope=-1.050 drop_kdac=-2.114
    row=29 col=14 noisy_transition=[36.0, 35.0] min_slope=-1.050 mean_slope=-1.041 drop_kdac=-2.098
    row=29 col=14 noisy_transition=[10.0, 9.0] min_slope=-1.032 mean_slope=-1.023 drop_kdac=-2.064
  Best bias = 304.0
  Max detectors in transition at best bias = 8
  Detectors at best bias in [224.0, 304.0]: [(8, 14), (12, 14), (13, 14), (14, 14), (15, 14), (23, 14), (24,
14), (27, 14)]
    row=8 col=14 transition=[266.0, 245.0] min_slope=-1.761 mean_slope=-0.354 drop_kdac=-7.785
    row=12 col=14 transition=[291.0, 254.0] min_slope=-3.500 mean_slope=-0.355 drop_kdac=-13.541
    row=13 col=14 transition=[256.0, 229.0] min_slope=-3.833 mean_slope=-0.479 drop_kdac=-13.415
    row=14 col=14 transition=[295.0, 273.0] min_slope=-2.775 mean_slope=-0.383 drop_kdac=-8.941
    row=15 col=14 transition=[291.0, 266.0] min_slope=-2.792 mean_slope=-0.392 drop_kdac=-10.226
    row=23 col=14 transition=[288.0, 264.0] min_slope=-2.687 mean_slope=-0.393 drop_kdac=-9.811
    row=24 col=14 transition=[245.0, 227.0] min_slope=-3.983 mean_slope=-0.579 drop_kdac=-10.947
    row=27 col=14 transition=[252.0, 224.0] min_slope=-3.059 mean_slope=-0.366 drop_kdac=-10.653
  User bias = 625
```

```
  Clean detectors in transition in [545.0, 625.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=1 col=14 transition=[643.0, 643.0] min_slope=-0.101 mean_slope=-0.101 drop_kdac=0.000
gap_to_window=18.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=22 col=14 transition=[654.0, 653.0] min_slope=-1.143 mean_slope=-1.091 drop_kdac=-2.286
gap_to_window=29.0 reason=excluded as noisy (multiple accepted transitions)
    row=26 col=14 transition=[574.0, 573.0] min_slope=-0.763 mean_slope=-0.756 drop_kdac=-1.524
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=14 transition=[540.0, 539.0] min_slope=-1.037 mean_slope=-1.036 drop_kdac=-2.072
gap_to_window=6.0 reason=excluded as noisy (multiple accepted transitions)
    row=31 col=14 transition=[572.0, 533.0] min_slope=-1.922 mean_slope=-0.253 drop_kdac=-10.137
gap_to_window=12.0 reason=accepted transition outside user window

Column 15
  Noisy detectors excluded: [(0, 15), (7, 15)]
    row=0 col=15 noisy_transition=[735.0, 732.0] min_slope=-1.349 mean_slope=-0.444 drop_kdac=-0.570
    row=0 col=15 noisy_transition=[686.0, 684.0] min_slope=-1.750 mean_slope=-0.823 drop_kdac=-0.873
    row=0 col=15 noisy_transition=[680.0, 670.0] min_slope=-1.669 mean_slope=-0.346 drop_kdac=-4.311
    row=0 col=15 noisy_transition=[657.0, 653.0] min_slope=-3.416 mean_slope=-1.072 drop_kdac=-2.549
    row=0 col=15 noisy_transition=[650.0, 648.0] min_slope=-1.343 mean_slope=-0.581 drop_kdac=-0.291
    row=0 col=15 noisy_transition=[645.0, 643.0] min_slope=-2.960 mean_slope=-1.910 drop_kdac=-5.919
    row=7 col=15 noisy_transition=[548.0, 543.0] min_slope=-1.283 mean_slope=-0.642 drop_kdac=-3.420
    row=7 col=15 noisy_transition=[540.0, 536.0] min_slope=-1.034 mean_slope=-0.775 drop_kdac=-3.483
  Best bias = 549.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [469.0, 549.0]: [(6, 15), (9, 15), (12, 15)]
    row=6 col=15 transition=[544.0, 534.0] min_slope=-3.813 mean_slope=-1.301 drop_kdac=-13.698
    row=9 col=15 transition=[521.0, 469.0] min_slope=-7.709 mean_slope=-0.524 drop_kdac=-27.772
    row=12 col=15 transition=[523.0, 473.0] min_slope=-6.925 mean_slope=-0.596 drop_kdac=-30.414
  User bias = 600
  Clean detectors in transition in [520.0, 600.0]: [(1, 15), (6, 15)]
    row=1 col=15 transition=[591.0, 535.0] min_slope=-6.980 mean_slope=-0.820 drop_kdac=-46.829
    row=6 col=15 transition=[544.0, 534.0] min_slope=-3.813 mean_slope=-1.301 drop_kdac=-13.698
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=7 col=15 transition=[548.0, 543.0] min_slope=-1.283 mean_slope=-0.642 drop_kdac=-3.420
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=7 col=15 transition=[540.0, 536.0] min_slope=-1.034 mean_slope=-0.775 drop_kdac=-3.483
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=15 transition=[554.0, 552.0] min_slope=-0.087 mean_slope=-0.082 drop_kdac=-0.154
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=15 transition=[549.0, 519.0] min_slope=-0.311 mean_slope=-0.161 drop_kdac=-4.765
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=15 transition=[578.0, 575.0] min_slope=-0.108 mean_slope=-0.090 drop_kdac=-0.299
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
```

```
    row=20 col=15 transition=[567.0, 526.0] min_slope=-0.882 mean_slope=-0.206 drop_kdac=-8.644
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 16
  Noisy detectors excluded: [(25, 16)]
    row=25 col=16 noisy_transition=[666.0, 665.0] min_slope=-1.124 mean_slope=-1.061 drop_kdac=-2.180
    row=25 col=16 noisy_transition=[657.0, 654.0] min_slope=-2.239 mean_slope=-1.161 drop_kdac=-4.690
    row=25 col=16 noisy_transition=[651.0, 648.0] min_slope=-1.602 mean_slope=-1.213 drop_kdac=-4.958
  Best bias = 487.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [407.0, 487.0]: [(3, 16), (12, 16)]
    row=3 col=16 transition=[441.0, 407.0] min_slope=-3.526 mean_slope=-0.381 drop_kdac=-13.304
    row=12 col=16 transition=[445.0, 444.0] min_slope=-1.344 mean_slope=-1.283 drop_kdac=-2.612
  User bias = 600
  Clean detectors in transition in [520.0, 600.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=20 col=16 transition=[639.0, 638.0] min_slope=-0.615 mean_slope=-0.606 drop_kdac=-1.206
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 17
  Noisy detectors excluded: [(8, 17)]
    row=8 col=17 noisy_transition=[256.0, 253.0] min_slope=-6.930 mean_slope=-3.469 drop_kdac=-14.105
    row=8 col=17 noisy_transition=[248.0, 245.0] min_slope=-8.165 mean_slope=-4.188 drop_kdac=-16.578
  Best bias = 488.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [408.0, 488.0]: [(19, 17)]
    row=19 col=17 transition=[436.0, 408.0] min_slope=-1.389 mean_slope=-0.205 drop_kdac=-6.027
  User bias = 500
  Clean detectors in transition in [420.0, 500.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=19 col=17 transition=[436.0, 408.0] min_slope=-1.389 mean_slope=-0.205 drop_kdac=-6.027
gap_to_window=12.0 reason=accepted transition outside user window
    row=23 col=17 transition=[447.0, 446.0] min_slope=-0.615 mean_slope=-0.605 drop_kdac=-1.209
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 18
  Noisy detectors excluded: [(13, 18), (14, 18), (18, 18)]
    row=13 col=18 noisy_transition=[1218.0, 1217.0] min_slope=-1.026 mean_slope=-0.985 drop_kdac=-2.047
    row=13 col=18 noisy_transition=[319.0, 294.0] min_slope=-1.165 mean_slope=-0.244 drop_kdac=-6.382
    row=14 col=18 noisy_transition=[604.0, 603.0] min_slope=-1.008 mean_slope=-0.955 drop_kdac=-2.017
    row=14 col=18 noisy_transition=[252.0, 238.0] min_slope=-1.351 mean_slope=-0.348 drop_kdac=-5.208
    row=18 col=18 noisy_transition=[1701.0, 1699.0] min_slope=-1.036 mean_slope=-0.688 drop_kdac=-2.073
    row=18 col=18 noisy_transition=[1532.0, 1531.0] min_slope=-1.026 mean_slope=-1.016 drop_kdac=-2.011
    row=18 col=18 noisy_transition=[1377.0, 1375.0] min_slope=-1.043 mean_slope=-0.690 drop_kdac=-2.086
    row=18 col=18 noisy_transition=[1229.0, 1228.0] min_slope=-1.028 mean_slope=-1.019 drop_kdac=-2.014
```

```
    row=18 col=18 noisy_transition=[1086.0, 1085.0] min_slope=-1.018 mean_slope=-1.008 drop_kdac=-1.993
    row=18 col=18 noisy_transition=[947.0, 946.0] min_slope=-1.030 mean_slope=-1.028 drop_kdac=-2.059
    row=18 col=18 noisy_transition=[808.0, 807.0] min_slope=-1.023 mean_slope=-1.013 drop_kdac=-2.005
    row=18 col=18 noisy_transition=[671.0, 669.0] min_slope=-1.041 mean_slope=-0.689 drop_kdac=-2.082
    row=18 col=18 noisy_transition=[533.0, 532.0] min_slope=-1.023 mean_slope=-1.006 drop_kdac=-1.977
    row=18 col=18 noisy_transition=[396.0, 395.0] min_slope=-1.026 mean_slope=-1.017 drop_kdac=-2.012
    row=18 col=18 noisy_transition=[256.0, 255.0] min_slope=-1.024 mean_slope=-1.014 drop_kdac=-2.007
    row=18 col=18 noisy_transition=[121.0, 120.0] min_slope=-1.024 mean_slope=-1.015 drop_kdac=-2.008
  Best bias = 358.0
  Max detectors in transition at best bias = 5
  Detectors at best bias in [278.0, 358.0]: [(7, 18), (8, 18), (19, 18), (24, 18), (28, 18)]
    row=7 col=18 transition=[347.0, 332.0] min_slope=-1.649 mean_slope=-0.357 drop_kdac=-5.713
    row=8 col=18 transition=[354.0, 333.0] min_slope=-1.442 mean_slope=-0.267 drop_kdac=-5.798
    row=19 col=18 transition=[302.0, 278.0] min_slope=-3.004 mean_slope=-0.412 drop_kdac=-10.286
    row=24 col=18 transition=[331.0, 314.0] min_slope=-1.667 mean_slope=-0.347 drop_kdac=-6.242
    row=28 col=18 transition=[320.0, 296.0] min_slope=-3.295 mean_slope=-0.419 drop_kdac=-10.493
  User bias = 375
  Clean detectors in transition in [295.0, 375.0]: [(7, 18), (8, 18), (24, 18), (28, 18)]
    row=7 col=18 transition=[347.0, 332.0] min_slope=-1.649 mean_slope=-0.357 drop_kdac=-5.713
    row=8 col=18 transition=[354.0, 333.0] min_slope=-1.442 mean_slope=-0.267 drop_kdac=-5.798
    row=24 col=18 transition=[331.0, 314.0] min_slope=-1.667 mean_slope=-0.347 drop_kdac=-6.242
    row=28 col=18 transition=[320.0, 296.0] min_slope=-3.295 mean_slope=-0.419 drop_kdac=-10.493
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=13 col=18 transition=[319.0, 294.0] min_slope=-1.165 mean_slope=-0.244 drop_kdac=-6.382
gap_to_window=1.0 reason=excluded as noisy (multiple accepted transitions)
    row=15 col=18 transition=[275.0, 258.0] min_slope=-1.187 mean_slope=-0.265 drop_kdac=-4.847
gap_to_window=37.0 reason=accepted transition outside user window
    row=18 col=18 transition=[396.0, 395.0] min_slope=-1.026 mean_slope=-1.017 drop_kdac=-2.012
gap_to_window=21.0 reason=excluded as noisy (multiple accepted transitions)
    row=18 col=18 transition=[256.0, 255.0] min_slope=-1.024 mean_slope=-1.014 drop_kdac=-2.007
gap_to_window=40.0 reason=excluded as noisy (multiple accepted transitions)
    row=19 col=18 transition=[302.0, 278.0] min_slope=-3.004 mean_slope=-0.412 drop_kdac=-10.286
gap_to_window=17.0 reason=accepted transition outside user window
    row=30 col=18 transition=[295.0, 273.0] min_slope=-4.413 mean_slope=-0.552 drop_kdac=-12.728
gap_to_window=22.0 reason=accepted transition outside user window

Column 19
  Noisy detectors excluded: []
  Best bias = 195.0
  Max detectors in transition at best bias = 6
  Detectors at best bias in [115.0, 195.0]: [(3, 19), (4, 19), (7, 19), (11, 19), (15, 19), (22, 19)]
    row=3 col=19 transition=[155.0, 137.0] min_slope=-2.189 mean_slope=-0.416 drop_kdac=-7.880
    row=4 col=19 transition=[180.0, 165.0] min_slope=-2.115 mean_slope=-0.420 drop_kdac=-6.722
    row=7 col=19 transition=[145.0, 129.0] min_slope=-2.639 mean_slope=-0.497 drop_kdac=-8.463
    row=11 col=19 transition=[195.0, 179.0] min_slope=-2.479 mean_slope=-0.456 drop_kdac=-7.701
```

```
    row=15 col=19 transition=[152.0, 135.0] min_slope=-2.667 mean_slope=-0.511 drop_kdac=-9.201
    row=22 col=19 transition=[125.0, 115.0] min_slope=-1.795 mean_slope=-0.458 drop_kdac=-5.138
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: [(11, 19), (14, 19), (24, 19)]
    row=11 col=19 transition=[195.0, 179.0] min_slope=-2.479 mean_slope=-0.456 drop_kdac=-7.701
    row=14 col=19 transition=[248.0, 222.0] min_slope=-3.413 mean_slope=-0.407 drop_kdac=-10.959
    row=24 col=19 transition=[243.0, 229.0] min_slope=-1.521 mean_slope=-0.351 drop_kdac=-5.338
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=19 transition=[155.0, 137.0] min_slope=-2.189 mean_slope=-0.416 drop_kdac=-7.880
gap_to_window=33.0 reason=accepted transition outside user window
    row=4 col=19 transition=[180.0, 165.0] min_slope=-2.115 mean_slope=-0.420 drop_kdac=-6.722
gap_to_window=5.0 reason=accepted transition outside user window
    row=5 col=19 transition=[282.0, 262.0] min_slope=-0.643 mean_slope=-0.199 drop_kdac=-4.205
gap_to_window=32.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=15 col=19 transition=[152.0, 135.0] min_slope=-2.667 mean_slope=-0.511 drop_kdac=-9.201
gap_to_window=35.0 reason=accepted transition outside user window
    row=20 col=19 transition=[266.0, 244.0] min_slope=-3.131 mean_slope=-0.422 drop_kdac=-9.657
gap_to_window=16.0 reason=accepted transition outside user window
    row=26 col=19 transition=[201.0, 185.0] min_slope=-0.919 mean_slope=-0.259 drop_kdac=-4.391
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=31 col=19 transition=[276.0, 253.0] min_slope=-2.696 mean_slope=-0.400 drop_kdac=-9.589
gap_to_window=26.0 reason=accepted transition outside user window

Column 20
  Noisy detectors excluded: [(15, 20)]
    row=15 col=20 noisy_transition=[269.0, 248.0] min_slope=-2.829 mean_slope=-0.401 drop_kdac=-8.950
    row=15 col=20 noisy_transition=[2.0, 1.0] min_slope=-1.155 mean_slope=-0.843 drop_kdac=-1.155
  Best bias = 278.0
  Max detectors in transition at best bias = 3
  Detectors at best bias in [198.0, 278.0]: [(5, 20), (9, 20), (13, 20)]
    row=5 col=20 transition=[278.0, 254.0] min_slope=-1.016 mean_slope=-0.220 drop_kdac=-5.586
    row=9 col=20 transition=[227.0, 207.0] min_slope=-2.142 mean_slope=-0.377 drop_kdac=-7.917
    row=13 col=20 transition=[216.0, 198.0] min_slope=-3.915 mean_slope=-0.597 drop_kdac=-11.366
  User bias = 500
  Clean detectors in transition in [420.0, 500.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=0 col=20 transition=[521.0, 520.0] min_slope=-0.586 mean_slope=-0.577 drop_kdac=-1.188
gap_to_window=21.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=0 col=20 transition=[472.0, 471.0] min_slope=-0.584 mean_slope=-0.575 drop_kdac=-1.185
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=0 col=20 transition=[423.0, 422.0] min_slope=-0.583 mean_slope=-0.575 drop_kdac=-1.184
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=25 col=20 transition=[440.0, 439.0] min_slope=-0.620 mean_slope=-0.611 drop_kdac=-1.221
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
```

```
Column 21
  Noisy detectors excluded: []
  Best bias = 280.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [200.0, 280.0]: [(11, 21), (12, 21)]
    row=11 col=21 transition=[240.0, 224.0] min_slope=-1.276 mean_slope=-0.264 drop_kdac=-4.455
    row=12 col=21 transition=[225.0, 200.0] min_slope=-2.283 mean_slope=-0.312 drop_kdac=-8.215
  User bias = 325
  Clean detectors in transition in [245.0, 325.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=6 col=21 transition=[300.0, 277.0] min_slope=-0.233 mean_slope=-0.127 drop_kdac=-2.980
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=11 col=21 transition=[240.0, 224.0] min_slope=-1.276 mean_slope=-0.264 drop_kdac=-4.455
gap_to_window=21.0 reason=accepted transition outside user window

Column 22
  Noisy detectors excluded: [(5, 22), (7, 22)]
    row=5 col=22 noisy_transition=[1483.0, 1482.0] min_slope=-1.089 mean_slope=-1.042 drop_kdac=-2.173
    row=5 col=22 noisy_transition=[870.0, 869.0] min_slope=-1.024 mean_slope=-1.005 drop_kdac=-2.044
    row=5 col=22 noisy_transition=[402.0, 401.0] min_slope=-1.038 mean_slope=-1.014 drop_kdac=-2.072
    row=7 col=22 noisy_transition=[814.0, 813.0] min_slope=-1.049 mean_slope=-0.992 drop_kdac=-2.094
    row=7 col=22 noisy_transition=[399.0, 398.0] min_slope=-1.015 mean_slope=-0.974 drop_kdac=-2.026
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 1100
  Clean detectors in transition in [1020.0, 1100.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=8 col=22 transition=[1136.0, 1135.0] min_slope=-0.720 mean_slope=-0.717 drop_kdac=-1.419
gap_to_window=36.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=8 col=22 transition=[1007.0, 1006.0] min_slope=-0.727 mean_slope=-0.720 drop_kdac=-1.443
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=10 col=22 transition=[1135.0, 1134.0] min_slope=-0.731 mean_slope=-0.728 drop_kdac=-1.449
gap_to_window=35.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=10 col=22 transition=[1014.0, 1013.0] min_slope=-0.726 mean_slope=-0.722 drop_kdac=-1.435
gap_to_window=7.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=24 col=22 transition=[1120.0, 1118.0] min_slope=-0.611 mean_slope=-0.403 drop_kdac=-1.221
gap_to_window=20.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 23
  Noisy detectors excluded: []
  Best bias = 308.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [228.0, 308.0]: [(4, 23)]
    row=4 col=23 transition=[246.0, 228.0] min_slope=-1.017 mean_slope=-0.236 drop_kdac=-4.407
```

```
  User bias = 325
  Clean detectors in transition in [245.0, 325.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=4 col=23 transition=[246.0, 228.0] min_slope=-1.017 mean_slope=-0.236 drop_kdac=-4.407
gap_to_window=17.0 reason=accepted transition outside user window
    row=5 col=23 transition=[260.0, 246.0] min_slope=-0.194 mean_slope=-0.119 drop_kdac=-1.653
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 24
  Noisy detectors excluded: []
  Best bias = 232.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [152.0, 232.0]: [(26, 24)]
    row=26 col=24 transition=[166.0, 152.0] min_slope=-2.713 mean_slope=-0.533 drop_kdac=-7.996
  User bias = 250
  Clean detectors in transition in [170.0, 250.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=26 col=24 transition=[166.0, 152.0] min_slope=-2.713 mean_slope=-0.533 drop_kdac=-7.996
gap_to_window=18.0 reason=accepted transition outside user window

Column 25
  Noisy detectors excluded: [(18, 25)]
    row=18 col=25 noisy_transition=[1679.0, 1676.0] min_slope=-3.340 mean_slope=-1.901 drop_kdac=-7.662
    row=18 col=25 noisy_transition=[1664.0, 1662.0] min_slope=-1.297 mean_slope=-0.867 drop_kdac=-2.594
    row=18 col=25 noisy_transition=[1653.0, 1650.0] min_slope=-3.013 mean_slope=-1.900 drop_kdac=-7.617
    row=18 col=25 noisy_transition=[1645.0, 1643.0] min_slope=-1.221 mean_slope=-0.830 drop_kdac=-2.442
    row=18 col=25 noisy_transition=[1634.0, 1632.0] min_slope=-1.235 mean_slope=-0.857 drop_kdac=-2.470
    row=18 col=25 noisy_transition=[1624.0, 1622.0] min_slope=-1.260 mean_slope=-0.842 drop_kdac=-2.520
    row=18 col=25 noisy_transition=[1619.0, 1617.0] min_slope=-1.223 mean_slope=-0.824 drop_kdac=-2.445
    row=18 col=25 noisy_transition=[689.0, 679.0] min_slope=-2.038 mean_slope=-0.922 drop_kdac=-10.098
  Best bias = 419.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [339.0, 419.0]: [(16, 25)]
    row=16 col=25 transition=[345.0, 339.0] min_slope=-1.112 mean_slope=-0.434 drop_kdac=-2.642
  User bias = 500
  Clean detectors in transition in [420.0, 500.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=14 col=25 transition=[397.0, 397.0] min_slope=-0.081 mean_slope=-0.081 drop_kdac=0.000
gap_to_window=23.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=17 col=25 transition=[471.0, 459.0] min_slope=-0.127 mean_slope=-0.107 drop_kdac=-1.305
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=19 col=25 transition=[392.0, 390.0] min_slope=-0.085 mean_slope=-0.082 drop_kdac=-0.153
gap_to_window=30.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=20 col=25 transition=[478.0, 476.0] min_slope=-0.183 mean_slope=-0.136 drop_kdac=-0.366
```

```
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 26
  Noisy detectors excluded: []
  Best bias = 347.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [267.0, 347.0]: [(1, 26)]
    row=1 col=26 transition=[289.0, 267.0] min_slope=-1.091 mean_slope=-0.279 drop_kdac=-6.378
  User bias = 400
  Clean detectors in transition in [320.0, 400.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=26 transition=[325.0, 305.0] min_slope=-0.200 mean_slope=-0.118 drop_kdac=-2.283
gap_to_window=15.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=26 transition=[307.0, 306.0] min_slope=-0.089 mean_slope=-0.087 drop_kdac=-0.170
gap_to_window=14.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=26 transition=[303.0, 301.0] min_slope=-0.084 mean_slope=-0.082 drop_kdac=-0.162
gap_to_window=19.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=7 col=26 transition=[298.0, 283.0] min_slope=-0.252 mean_slope=-0.139 drop_kdac=-1.931
gap_to_window=37.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=12 col=26 transition=[409.0, 409.0] min_slope=-0.132 mean_slope=-0.132 drop_kdac=0.000
gap_to_window=9.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[401.0, 401.0] min_slope=-0.104 mean_slope=-0.104 drop_kdac=0.000
gap_to_window=1.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=12 col=26 transition=[313.0, 312.0] min_slope=-0.247 mean_slope=-0.219 drop_kdac=-0.468
gap_to_window=8.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=26 transition=[395.0, 395.0] min_slope=-0.109 mean_slope=-0.109 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[389.0, 389.0] min_slope=-0.219 mean_slope=-0.219 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[384.0, 384.0] min_slope=-0.135 mean_slope=-0.135 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[378.0, 378.0] min_slope=-0.082 mean_slope=-0.082 drop_kdac=0.000
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[319.0, 317.0] min_slope=-0.118 mean_slope=-0.102 drop_kdac=-0.192
gap_to_window=3.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=13 col=26 transition=[311.0, 311.0] min_slope=-0.255 mean_slope=-0.255 drop_kdac=0.000
gap_to_window=9.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[307.0, 307.0] min_slope=-0.091 mean_slope=-0.091 drop_kdac=0.000
```

```
gap_to_window=13.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule
    row=13 col=26 transition=[281.0, 281.0] min_slope=-0.189 mean_slope=-0.189 drop_kdac=0.000
gap_to_window=39.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 27
  Noisy detectors excluded: [(8, 27), (22, 27)]
    row=8 col=27 noisy_transition=[752.0, 740.0] min_slope=-2.352 mean_slope=-0.593 drop_kdac=-7.618
    row=8 col=27 noisy_transition=[737.0, 734.0] min_slope=-1.676 mean_slope=-0.963 drop_kdac=-4.254
    row=8 col=27 noisy_transition=[669.0, 668.0] min_slope=-1.371 mean_slope=-1.289 drop_kdac=-2.614
    row=8 col=27 noisy_transition=[665.0, 662.0] min_slope=-1.381 mean_slope=-0.707 drop_kdac=-2.958
    row=8 col=27 noisy_transition=[619.0, 617.0] min_slope=-1.304 mean_slope=-0.854 drop_kdac=-2.608
    row=8 col=27 noisy_transition=[572.0, 567.0] min_slope=-3.805 mean_slope=-1.932 drop_kdac=-11.599
    row=8 col=27 noisy_transition=[504.0, 503.0] min_slope=-1.494 mean_slope=-1.390 drop_kdac=-2.964
    row=8 col=27 noisy_transition=[385.0, 384.0] min_slope=-1.183 mean_slope=-1.077 drop_kdac=-2.419
    row=22 col=27 noisy_transition=[528.0, 491.0] min_slope=-3.720 mean_slope=-0.441 drop_kdac=-15.515
    row=22 col=27 noisy_transition=[484.0, 479.0] min_slope=-1.641 mean_slope=-0.859 drop_kdac=-4.932
    row=22 col=27 noisy_transition=[476.0, 470.0] min_slope=-2.366 mean_slope=-0.751 drop_kdac=-5.231
  Best bias = 650.0
  Max detectors in transition at best bias = 2
  Detectors at best bias in [570.0, 650.0]: [(15, 27), (23, 27)]
    row=15 col=27 transition=[610.0, 577.0] min_slope=-3.578 mean_slope=-0.613 drop_kdac=-20.782
    row=23 col=27 transition=[610.0, 570.0] min_slope=-1.026 mean_slope=-0.171 drop_kdac=-7.095
  User bias = 750
  Clean detectors in transition in [670.0, 750.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=8 col=27 transition=[752.0, 740.0] min_slope=-2.352 mean_slope=-0.593 drop_kdac=-7.618
gap_to_window=2.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=27 transition=[737.0, 734.0] min_slope=-1.676 mean_slope=-0.963 drop_kdac=-4.254
gap_to_window=0.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=27 transition=[669.0, 668.0] min_slope=-1.371 mean_slope=-1.289 drop_kdac=-2.614
gap_to_window=2.0 reason=excluded as noisy (multiple accepted transitions)
    row=8 col=27 transition=[665.0, 662.0] min_slope=-1.381 mean_slope=-0.707 drop_kdac=-2.958
gap_to_window=8.0 reason=excluded as noisy (multiple accepted transitions)
    row=9 col=27 transition=[674.0, 673.0] min_slope=-0.094 mean_slope=-0.092 drop_kdac=-0.149
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=26 col=27 transition=[668.0, 666.0] min_slope=-1.179 mean_slope=-0.777 drop_kdac=-2.358
gap_to_window=4.0 reason=accepted transition outside user window
    row=27 col=27 transition=[667.0, 667.0] min_slope=-0.085 mean_slope=-0.085 drop_kdac=0.000
gap_to_window=3.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule,
segment too short for sustained rule

Column 28
  Noisy detectors excluded: []
```

```
  Best bias = 713.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [633.0, 713.0]: [(26, 28)]
    row=26 col=28 transition=[655.0, 633.0] min_slope=-3.354 mean_slope=-0.398 drop_kdac=-9.164
  User bias = 750
  Clean detectors in transition in [670.0, 750.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=26 col=28 transition=[655.0, 633.0] min_slope=-3.354 mean_slope=-0.398 drop_kdac=-9.164
gap_to_window=37.0 reason=accepted transition outside user window

Column 29
  Noisy detectors excluded: []
  Best bias = 667.0
  Max detectors in transition at best bias = 1
  Detectors at best bias in [587.0, 667.0]: [(3, 29)]
    row=3 col=29 transition=[596.0, 587.0] min_slope=-1.566 mean_slope=-0.393 drop_kdac=-3.912
  User bias = 700
  Clean detectors in transition in [620.0, 700.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    row=3 col=29 transition=[596.0, 587.0] min_slope=-1.566 mean_slope=-0.393 drop_kdac=-3.912
gap_to_window=33.0 reason=accepted transition outside user window
    row=17 col=29 transition=[671.0, 670.0] min_slope=-0.141 mean_slope=-0.129 drop_kdac=-0.238
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[643.0, 642.0] min_slope=-0.156 mean_slope=-0.120 drop_kdac=-0.166
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=17 col=29 transition=[614.0, 613.0] min_slope=-0.152 mean_slope=-0.139 drop_kdac=-0.252
gap_to_window=7.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[724.0, 723.0] min_slope=-0.164 mean_slope=-0.132 drop_kdac=-0.282
gap_to_window=24.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[696.0, 695.0] min_slope=-0.202 mean_slope=-0.175 drop_kdac=-0.373
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[667.0, 666.0] min_slope=-0.186 mean_slope=-0.158 drop_kdac=-0.338
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[638.0, 637.0] min_slope=-0.171 mean_slope=-0.142 drop_kdac=-0.307
gap_to_window=0.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[610.0, 609.0] min_slope=-0.215 mean_slope=-0.189 drop_kdac=-0.401
gap_to_window=11.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule
    row=29 col=29 transition=[581.0, 580.0] min_slope=-0.201 mean_slope=-0.173 drop_kdac=-0.368
gap_to_window=40.0 reason=slope not steep enough for sharp rule, total drop too small for sustained rule

Column 30
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
```

```
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Column 31
  Noisy detectors excluded: []
  Best bias = 2000.0
  Max detectors in transition at best bias = 0
  Detectors at best bias in [1920.0, 2000.0]: []
  User bias = 0
  Clean detectors in transition in [-80.0, 0.0]: []
  Near misses / diagnostics (<=40 DAC from window, or rejected/noisy overlaps):
    []

Saved PDF: /home/timestudent/anie/load_curves_all/specB/best_bias_20260210_spec_B_iv_344mK_iv_all_data.pdf
```